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Symbolic Versus Substantive ESG Practices: A Systematic Review and Integrative Framework

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Correspondence: Cristina Alexandrina Ștefănescu (cristina.palfi@econ.ubbcluj.ro)**Received:** 9 January 2026 | **Revised:** 15 April 2026 | **Accepted:** 20 May 2026**Keywords:** conceptual framework | consequences | determinants | ESG decoupling | systematic literature review

ABSTRACT

ESG reporting is widespread, but symbolic commitments do not always reflect substantive practices. This study conducts a systematic literature review of 62 empirical articles published between 2021 and 2025 to synthesize the main determinants and consequences of this disclosure–performance misalignment in ESG reporting (commonly referred to as ESG decoupling). Following the SPAR-4-SLR protocol and applying the TCCM framework, the review organizes evidence across theories, contexts, characteristics, and methodological approaches. Drawing on systems theory and performance feedback logic, the study develops an integrative conceptual framework explaining how symbolic and substantive ESG practices interact over time. The framework identifies four recurring mechanism clusters—agency and governance, institutional and regulatory, information asymmetry and signaling, and resource and capability—that shape short-term tactical responses and longer-term structural misalignment. The review provides a coherent, theory-informed synthesis and offers structured insights for future research and for policymakers, managers, and investors concerned with the credibility of ESG reporting.

JEL Classification: Q5, Q01

1 | Introduction

Over the past decade, corporate sustainability has increasingly been framed through the lens of Environmental, Social, and Governance (ESG) accountability, as firms face mounting pressure from regulators, investors, and society to justify not only what they report but also how sustainability commitments are enacted in practice (Aboud et al. 2024; Arena et al. 2025). ESG disclosure has evolved from a voluntary signaling device into a core governance mechanism, particularly in jurisdictions where reporting obligations are expanding, and external scrutiny is intensifying (Li et al. 2025a; Chen et al. 2025). This shift has elevated expectations regarding the credibility and decision usefulness of ESG information, positioning disclosure as a key interface between firms and stakeholders (Talpur et al. 2024; Cepêda et al. 2025b).

High-profile corporate controversies illustrate ESG decoupling as the misalignment between sustainability disclosures and underlying practices. The Volkswagen emissions scandal, for example, shows how strong environmental positioning can coexist with practices that contradict stated commitments (Zhang et al. 2021; Boiral et al. 2021). Similar divergences appear in the Deepwater Horizon case (Matejek and Gössling 2014) and in governance-related misconduct at Wells Fargo (Noel and Osman 2024). More recent evidence reinforces this pattern, as regulatory investigations into DWS Group have raised concerns about overstated ESG integration, highlighting the gap between ESG claims and their implementation (Huang et al. 2024a). Together, these cases suggest that ESG disclosure may function as a symbolic rather than substantive indicator of sustainability performance.

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Study	Type of review	Core focus	Main contribution
Cepêda et al. (2025)	Bibliometric and thematic literature review	Sustainability reporting decoupling	Maps publication trends, theories, proxies, and identifies five research streams in decoupling literature
Talpur et al. (2023)	Systematic narrative literature review (PRISMA-based)	CSR decoupling drivers and consequences	Provides a broad thematic synthesis of CSR decoupling and outlines a future research agenda
Luan (2024)	Context-focused narrative review	CSR decoupling across ESG pillars	Synthesizes CSR decoupling literature with a focus on environmental, social, and governance dimensions
Loko and Schiehl (2025)	Conceptual framework with empirical validation	ESG policy–practice decoupling	Develops and empirically validates a granular measurement framework for ESG misalignment
Khan and Lockhart (2022)	Conceptual and theory-driven review	CSR decoupling in developing contexts	Re-theorizes CSR decoupling using institutional theory and proposes a context-sensitive research agenda
Velte (2023)	Narrative review of empirical quantitative studies	Determinants and consequences of CSR decoupling	Synthesizes firm- and country-level determinants and highlights measurement and methodological limitations
Vadén et al. (2020)	Conceptual review and typology development	Environmental decoupling at macro level	Develops a categorization of decoupling types relevant for ecological sustainability
<i>This study</i>	<i>Theory-driven integrative literature review with framework development</i>	<i>ESG decoupling as a dynamic organizational process</i>	<i>Develops an integrative framework linking determinants and consequences through core mechanisms, temporal patterns, and feedback loops</i>

FIGURE 1 | Summary of prior review studies.

These concerns are also reflected in the academic literature. A growing body of evidence shows that more extensive disclosure does not necessarily translate into improved sustainability performance (Li et al. 2025a; He et al. 2024). Firms may formally comply with ESG requirements while maintaining practices that fall short of stated commitments, raising concerns about the substantive effectiveness of ESG reporting regimes (Velte 2023; Carè et al. 2025; Zhang and Xiong 2025). This persistent misalignment between ESG rhetoric and organizational practice—hereafter referred to as ESG decoupling—has prompted renewed academic attention to the question of when ESG disclosure enhances accountability and when it obscures underlying inconsistencies (Talpur et al. 2024; Cepêda et al. 2025a).

Prior research identifies a combination of organizational and institutional conditions that shape the emergence of ESG decoupling. On the external side, weak regulatory enforcement, fragmented reporting regimes, and limited assurance allow firms to satisfy formal disclosure requirements without substantively changing ESG practices, thereby increasing reliance on symbolic compliance (Aboud et al. 2024; Seow 2023; Arena et al. 2025). Internally, governance quality plays a critical role. Weak board oversight, limited independence, and concentrated decision-making authority amplify managerial discretion, enabling selective transparency and strategic ESG disclosure (Disli et al. 2022; Eliwa et al. 2023; Zhao et al. 2024). By contrast, stronger governance arrangements—such as more independent and diverse boards, enhanced monitoring structures, and greater use of digital technologies—improve traceability and constrain opportunistic reporting behavior, thereby reducing the scope for sustained ESG decoupling (Eliwa et al. 2023; Li et al. 2024b; Liu, Lu, Zhao, et al. 2024d).

The consequences of ESG decoupling extend well beyond reporting credibility. While symbolic ESG disclosure may generate

short-term benefits, including favorable investor reactions or temporary financing advantages, these effects tend to dissipate as inconsistencies between disclosure and performance become more visible (Candelon et al. 2021; Wang et al. 2025). Over time, ESG decoupling undermines organizational legitimacy, erodes stakeholder trust, and diminishes the informational value of sustainability reporting (De La Fuente and Velasco 2024; Yang et al. 2024). Empirical evidence further links persistent decoupling to reduced innovation efficiency, resource misallocation, higher capital costs, and long-term value deterioration (Di and Li 2023; Sun et al. 2025b; Hussain et al. 2025). Taken together, these findings suggest that although ESG decoupling may offer short-lived strategic relief, it ultimately exposes firms to cumulative reputational, operational, and financial risks (Venturelli et al. 2024; García-Meca et al. 2025).

Despite its growing relevance, ESG decoupling has not yet been the subject of a comprehensive, theory-informed synthesis. Existing review studies largely address related but conceptually distinct domains (see Figure 1).

Some studies provide bibliometric or thematic mappings of sustainability reporting decoupling without developing explanatory frameworks or identifying underlying causal mechanisms (Cepêda et al. 2025b). Others examine the drivers and outcomes of CSR decoupling using largely static classifications, such as internal versus external determinants or positive versus negative effects (Talpur et al. 2024; Velte 2023), or apply theoretical lenses in isolation rather than in an integrated manner (Hahn and Kühnen 2013; Seow 2023; Arkoh et al. 2024). Additional studies focus on specific institutional contexts, including developing economies or single-country settings (Khan and Lockhart 2022; Luan 2024), or refine measurement approaches for policy–practice misalignment without theorizing

ESG decoupling as an evolving organizational process (Loko and Schiehl 2025).

While these studies provide valuable insights, they offer only a limited understanding of how and why ESG decoupling emerges, persists, and is corrected over time, and do not provide a unifying framework that connects its determinants and consequences within a dynamic, theory-informed perspective.

To address these gaps, this study aims to develop a theory-informed and integrative understanding of ESG decoupling as a dynamic organizational phenomenon rather than a static reporting outcome. To achieve this overarching objective, the research pursues three interrelated steps. First, it systematically synthesizes the empirical literature on ESG decoupling, focusing on identifying its key determinants, consequences, and underlying theoretical explanations. Second, it organizes this body of evidence using the TCCM framework, enabling a structured assessment of how ESG decoupling has been conceptualized and empirically investigated across different contexts and methodological approaches. Third, it develops an integrative conceptual framework that captures the interplay among underlying mechanisms, temporal dynamics, and feedback processes shaping ESG decoupling over time.

Methodologically, the review follows the SPAR-4-SLR protocol (Paul et al. 2021), ensuring transparency, replicability, and analytical rigor in the identification, screening, and synthesis of prior studies. To structure the analysis, the study adopts the TCCM framework (Theory–Context–Characteristics–Methodology) (Paul et al. 2024; Mukhtar et al. 2025), which enables a systematic assessment of the theoretical perspectives, empirical contexts, conceptualizations, and methodological approaches employed in the ESG decoupling literature.

Building on this structured synthesis of 62 empirical studies published between 2021 and 2025, the review develops an integrative conceptual framework that links the determinants and consequences of ESG decoupling through four core mechanisms: agency and governance; institutional and regulatory; information asymmetry and signaling; and resource and capability. Drawing on systems theory (Bertalanffy 1968; Leoni 2024) and performance feedback logic (Cyert and March 1992; Greve 2003), the framework conceptualizes ESG decoupling as an adaptive organizational process shaped by interactions among internal incentives, external pressures, and feedback from markets and stakeholders. Accordingly, the framework incorporates a temporal dimension—distinguishing between short-term tactical decoupling and long-term structural misalignment—and introduces feedback loops that explain how the consequences of symbolic ESG behavior may either reinforce or correct subsequent organizational actions.

The contributions of this review are threefold. First, at the theoretical level, the study advances ESG research by shifting the analytical focus from static, fragmented explanations to a process-oriented understanding of ESG decoupling, emphasizing its dynamic, adaptive nature. Second, at the conceptual and empirical level, the study provides an integrated synthesis that connects previously isolated findings on determinants and consequences, offering a more coherent explanation of the

conditions under which ESG decoupling persists, intensifies, or is corrected across different contexts. Third, at the practical level, the proposed framework offers actionable insights for regulators, investors, and corporate decision-makers by identifying the underlying mechanisms driving ESG misalignment and clarifying when governance, regulatory, informational, or capability-based interventions are most likely to enhance the credibility of ESG commitments.

The remainder of the article is structured as follows: Section 2 describes the systematic review methodology. Section 3 presents the findings, organized by the TCCM framework. Section 4 introduces the integrative conceptual framework. Section 5 discusses future research directions, and Section 6 concludes with a summary of key theoretical contributions, practical implications, and limitations.

2 | Review Methodology

This study employs a systematic literature review (SLR) to examine the determinants and consequences of ESG decoupling and to develop an integrative, theory-based framework that captures its underlying mechanisms, contextual influences, and temporal dynamics. An SLR is particularly suitable for this purpose, as research on ESG decoupling spans diverse theoretical perspectives, empirical contexts, and methodological approaches, requiring a structured and transparent synthesis to enable cumulative knowledge development. While related reviews have examined decoupling in CSR or sustainability reporting (e.g., Talpur et al. 2024; Velte 2023), this review focuses explicitly on empirical studies examining ESG-related disclosure–performance misalignment.

Following established guidelines for systematic reviews in management and sustainability research (Tranfield et al. 2003; Snyder 2019), the review applies explicit inclusion criteria and a concept-driven synthesis to ensure rigor and replicability. Rather than relying on descriptive narrative approaches or citation-based techniques (Massaro et al. 2016; Farrukh et al. 2020), the SLR emphasizes theoretical integration and empirical convergence (Paul and Criado 2020), thereby supporting conceptual clarity and critical evaluation across studies (Snyder 2019; Torraco 2016).

To structure the analysis, the review adopts the TCCM framework (Paul et al. 2024; Mukhtar et al. 2025) to structure the review around four key dimensions: Theories (T) used, Contexts (C) studied, Characteristics (C) of key constructs, and Methodologies (M) employed. Compared with purely descriptive (Massaro et al. 2016) or bibliometric reviews (Farrukh et al. 2020; Zhang and Xiong 2025), the TCCM framework facilitates deeper analytical integration and the identification of underexplored theoretical and contextual intersections (Sharma et al. 2020).

To ensure procedural rigor and replicability (Palmatier et al. 2018), the review follows the SPAR-4-SLR protocol (Paul et al. 2021), which involves three main stages: (1) assembling, (2) arranging, and (3) assessing. As shown in Figure 2, this step-by-step process provides a clear foundation for selecting, organizing, and interpreting relevant literature. Together,

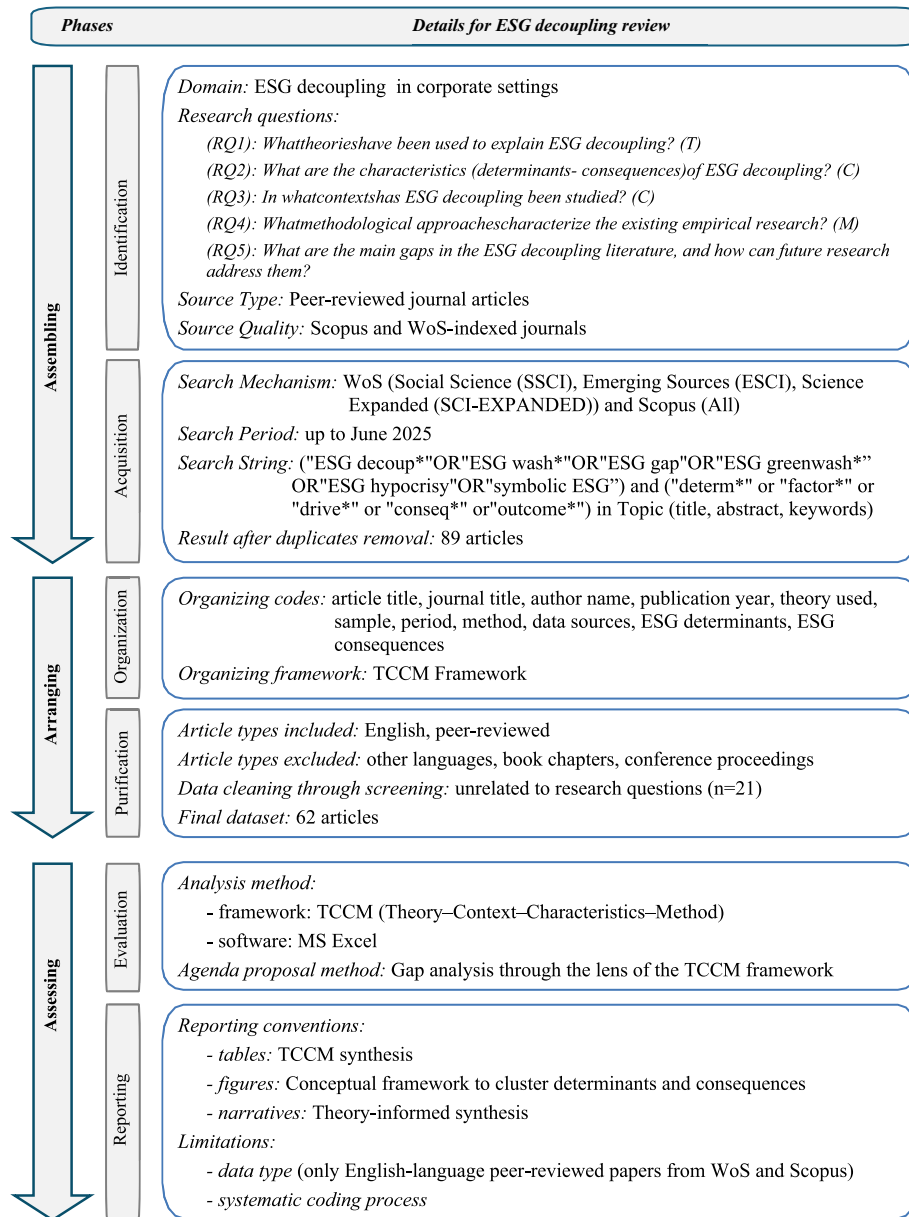


FIGURE 2 | SPAR-4-SLR flowchart.

the SPAR-4-SLR protocol and the TCCM framework provide a coherent methodological foundation for systematically identifying, organizing, and synthesizing empirical research on ESG decoupling.

2.1 | Assembling

The first stage of the SPAR-4-SLR protocol is assembling, which comprises two sub-stages: identification and acquisition (Paul et al. 2021).

The *identification phase* focused on clearly defining the scope and analytical orientation of the review and on specifying the guiding research questions.

Given the diversity of theoretical perspectives, empirical settings, and measurement approaches observed in the ESG decoupling

literature, a question-driven structure was adopted to ensure alignment with the TCCM framework and to facilitate systematic synthesis across studies. Accordingly, the review was guided by the following research questions:

RQ1. What theories have been used to explain ESG decoupling? (T).

RQ2. What are the key characteristics of ESG decoupling in terms of its determinants and consequences? (C).

RQ3. In what contexts (e.g., countries, industries, institutional settings) has ESG decoupling been studied? (C).

RQ4. What methodological approaches characterize the existing empirical research on ESG decoupling? (M).

RQ5. *What gaps remain in the ESG decoupling literature, and what future research directions can advance understanding in this domain?*

These research questions provided a structured basis for identifying relevant studies and ensured consistency between the review's objectives, the TCCM framework, and the subsequent stages of analysis.

The *acquisition phase* involved a thorough search across two major academic databases: Web of Science (WoS) and Scopus. These databases were selected due to their extensive coverage of peer-reviewed literature in business, management, and sustainability research (Falagas et al. 2008). WoS provides curated citation indexing that facilitates the identification of influential studies and the assessment of intellectual structure (Wang and Waltman 2016), while Scopus offers broader disciplinary coverage, particularly across management and sustainability journals (Baas et al. 2020).

The search strategy combined a set of keywords capturing ESG decoupling and related disclosure–performance misalignment concepts with terms reflecting their antecedents and outcomes. Specifically, the search string included (“ESG decoup*” OR “ESG wash*” OR “ESG gap” OR “ESG green-wash*” OR “ESG hypocrisy” OR “symbolic ESG”) AND (“de-term*” OR “factor*” OR “driver*” OR “conseq*” OR “out-come*”). Boolean operators (AND, OR) were applied across titles, abstracts, and keywords (topic search) in both Web of Science and Scopus to identify empirical studies that explicitly examine ESG decoupling and related phenomena. The full search string is presented in Figure 2 to ensure transparency and replicability. To enhance coverage, backward and forward reference checks were conducted on the final set of eligible articles.

Following the removal of duplicate records, 89 unique studies were retained for subsequent relevance screening and full-text assessment.

2.2 | Arranging

The arranging stage involved converting the initial pool of studies collected during the *assembling phase* into a structured dataset suitable for conceptual and theoretical synthesis. This stage combined systematic organization with rigorous filtering to ensure analytical consistency and transparency throughout the review process.

Each article was coded using a standardized Excel-based matrix aligned with the TCCM framework (Paul and Rosado-Serrano 2019). The coding scheme included the following elements: article title; author(s); publication year; theoretical lens; empirical context and sample characteristics; study period; methodological approach; data sources; and the determinants and consequences of ESG decoupling examined. These coding categories correspond to the four dimensions of the TCCM framework—Theories (T), Contexts (C), Characteristics (C), and Methods (M)—and enable systematic comparison across studies.

To ensure a transparent and consistent selection process, predefined inclusion criteria were applied. Studies were required to explicitly examine ESG decoupling or closely related disclosure–performance misalignment concepts as a central focus of the empirical analysis. The screening followed a two-step procedure. First, only empirical studies were retained to ensure an evidence-based dataset. Second, a full-text assessment was conducted to verify each study's alignment with the review's scope and analytical focus.

To further refine the dataset, a structured *purification stage* was implemented by retaining only peer-reviewed journal articles published in English, including those available as early access or in press, in line with established systematic review guidelines (Tranfield et al. 2003; Snyder 2019). Although no restrictions were imposed during the initial search regarding journal type or publication year, the final sample comprises 62 empirical articles published between 2021 and 2025 (see Table B1, Appendix B). This period reflects the recent emergence of ESG decoupling as a distinct focus of empirical inquiry. The resulting dataset provided a coherent and thematically consistent foundation for subsequent theory-informed assessment and synthesis.

2.3 | Assessing

The *evaluation stage* involved a systematic assessment of the reviewed studies across the four dimensions of the TCCM framework (Paul et al. 2024): theoretical foundations; empirical contexts (e.g., regions and sectors); core characteristics of ESG decoupling (including determinants and consequences); and methodological approaches.

To synthesize the literature systematically and in a theory-informed manner, the review employed Mayring's (2000) qualitative content analysis. This approach enabled the systematic identification and comparison of key concepts and relationships across studies, supporting the identification of recurring patterns and shared causal logics.

The analytical process followed a structured, multi-step procedure. First, individual studies were coded along the TCCM dimensions, with particular attention to how ESG decoupling, its determinants, and its consequences were operationalized. Second, recurring patterns were identified by comparing studies, focusing on theoretical explanations, empirical contexts, and observed relationships between determinants and outcomes. Third, these patterns were synthesized into mechanisms grounded on shared causal logic that links determinants and consequences.

Consistent with integrative review principles (Torraco 2016), this approach facilitated the transition from descriptive aggregation to theory-informed synthesis and provided the analytical foundation for developing the integrative conceptual framework. In coding the characteristics of ESG decoupling, particular attention was paid to how studies operationalize disclosure–performance misalignment. Determinants and consequences were identified based on the explicit empirical relationships tested in each study, rather than inferred associations, thereby reducing subjectivity in categorization.

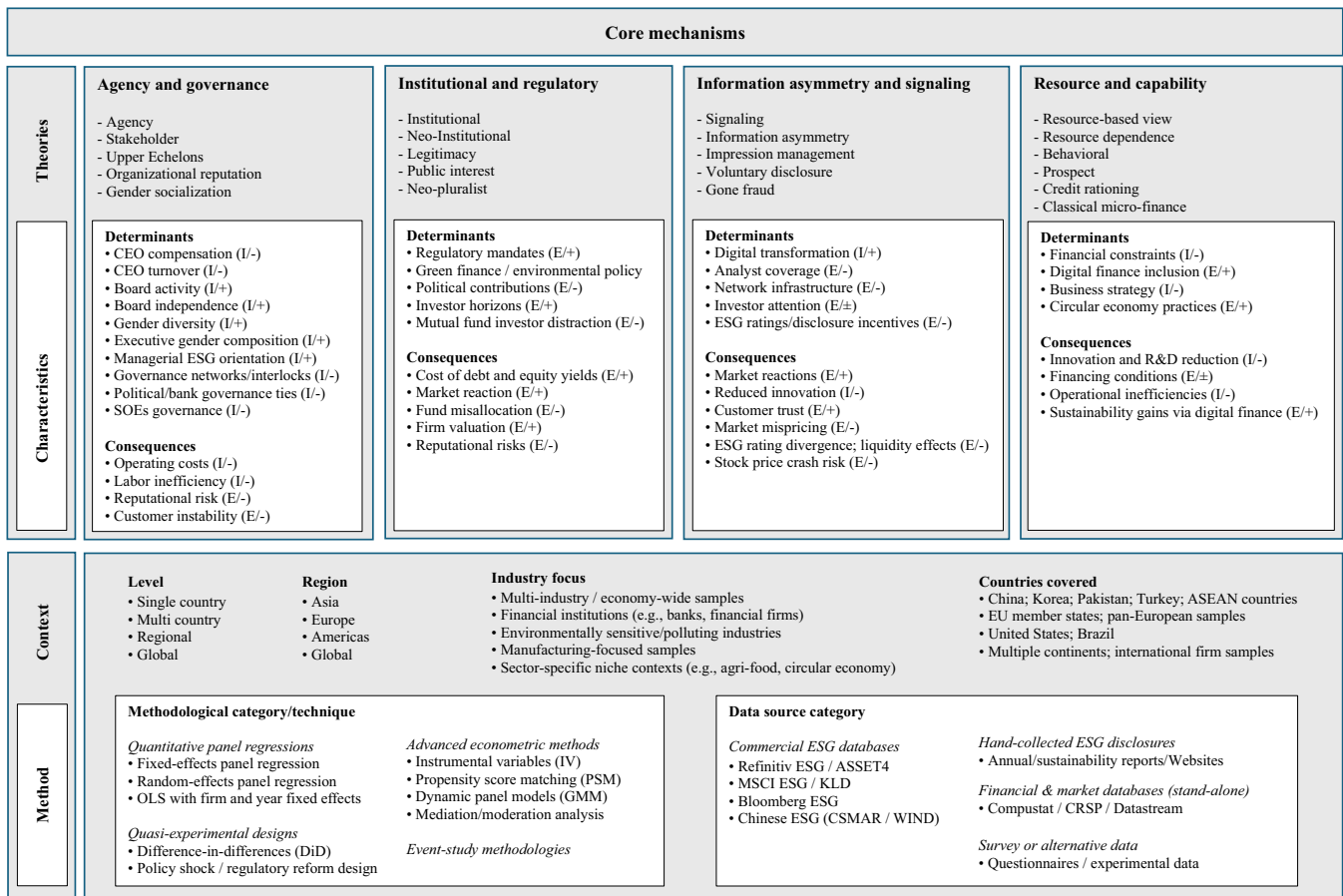


FIGURE 3 | TCCM framework of ESG decoupling.

The *presentation* of findings followed a multi-format synthesis strategy, combining tables summarizing dominant patterns, frequencies, and distributions with narrative analysis highlighting conceptual linkages and empirical trends.

While the SPAR-4-SLR protocol provides a transparent and replicable process, a few limitations must be acknowledged. First, the literature may be subject to publication bias, particularly the tendency to report statistically significant or confirmatory results (Pickering and Byrne 2014). Second, although systematic coding procedures were applied, the interpretation of constructs such as antecedents and outcomes necessarily involves subjective judgment. Accordingly, the resulting framework should be viewed as analytically grounded yet provisional, providing a foundation for future empirical refinement and theory development.

3 | Review Findings

This section synthesizes the review findings to examine how ESG decoupling has been analyzed across theoretical perspectives, contextual settings, core characteristics, and methodological approaches, following the TCCM framework (see Figure 3).

3.1 | Theory (T)

The ESG decoupling literature builds on a broad but uneven set of theoretical perspectives to explain why ESG disclosure diverges

from underlying practices (see Appendix A—Table A1). A limited number of dominant frameworks—particularly agency, stakeholder, and signaling theories—account for most explanations, while other perspectives are applied less frequently and often in isolation. As a result, the literature remains theoretically fragmented, with studies offering partial and sometimes disconnected explanations of ESG decoupling, thereby posing challenges for synthesis, as differences in theoretical framing may reduce the comparability and integration of findings.

The most influential explanations are rooted in managerial and governance-oriented theories. Agency theory is the dominant lens, framing ESG decoupling as a consequence of managerial self-interest, short-term incentives, and weak monitoring, which encourage symbolic ESG disclosure rather than substantive change (Cepêda et al. 2025a; Li et al. 2024a; Disli et al. 2022; Zhao et al. 2024). Upper echelons theory complements this view by emphasizing how executive characteristics—such as gender composition or leadership turnover—shape the authenticity of ESG engagement (Eliwa et al. 2023; Niu et al. 2024; Yang et al. 2024). Together, these perspectives locate the origins of ESG decoupling primarily within the firm, highlighting the role of managerial discretion, incentives, and governance structures.

A second stream of research relies on theories that emphasize external pressures and legitimacy concerns. Institutional and stakeholder perspectives explain ESG decoupling as symbolic conformity to regulatory requirements and stakeholder

expectations when enforcement is weak or compliance is costly (Aboud et al. 2024; Li et al. 2024a; Venturelli et al. 2024). Relatedly, legitimacy and resource dependence theories highlight how firms use ESG disclosure to preserve social acceptance or secure access to capital and political support despite limited substantive ESG performance (Carè et al. 2025; Liu et al. 2024b; Yin et al. 2024). In contrast to governance-based explanations, these perspectives shift the focus from internal incentives to external pressures, suggesting that ESG decoupling may also represent an adaptive response to institutional complexity and stakeholder expectations.

A third group of studies focuses on information and communication dynamics. Signaling theory portrays ESG disclosure as a tool for sending favorable signals to capital markets even when underlying practices do not align (Candelon et al. 2021; Liao et al. 2023; Tao et al. 2024). Information asymmetry and impression management perspectives further argue that managers exploit disclosure gaps, ESG ratings, and reporting flexibility to shape stakeholder perceptions and obscure weak ESG performance (Chen et al. 2024a; Lin et al. 2023; Hussain et al. 2025). These perspectives complement both governance- and institution-based explanations by emphasizing the strategic role of disclosure itself, framing ESG decoupling as a communication-driven phenomenon shaped by information asymmetries and market expectations.

Finally, a smaller but relevant set of studies adopts resource- and behavior-oriented perspectives. These approaches explain ESG decoupling as an adaptive response to financial constraints, capability gaps, or performance pressures, in which symbolic ESG disclosure serves as a low-cost alternative to substantive sustainability investments (Zhang 2022; Li and Chen 2024; Kweh et al. 2026). Taken together, these perspectives highlight that ESG decoupling may also arise from structural and capability-related constraints, rather than solely from opportunistic or legitimacy-driven behavior. Notably, a considerable share of the literature combines multiple theories or employs no explicit theoretical framework, often applying them in an isolated or loosely connected manner, underscoring both the fragmentation of existing explanations and the need for a more integrative perspective.

3.2 | Context (C)

The empirical literature on ESG decoupling is highly context-concentrated, both geographically and industrially (see Appendix A—Table A2).

Single-country studies account for 67.7% of the sample, with China alone accounting for 59.7% of the empirical evidence. This dominance reflects the salience of regulatory reforms, state ownership, and governance restructuring in shaping ESG disclosure–performance misalignments in the Chinese context (Niu et al. 2024; Zhao et al. 2024; Chen et al. 2024a; Wang et al. 2025a). While this setting enables detailed institutional analysis, it also embeds much of the existing evidence within a specific emerging-market governance environment, potentially limiting its transferability to other institutional settings (Li et al. 2024a; Jin et al. 2024).

By contrast, regional studies account for only 9.7%, primarily focusing on Europe and the EU, where ESG decoupling is examined through the lens of regulatory harmonization, stakeholder protection, and disclosure enforcement (Disli et al. 2022; Eliwa et al. 2023; Aboud et al. 2024; Venturelli et al. 2024). Global and multi-country studies represent 22.6% of the sample and provide comparative insights into how ESG decoupling affects valuation, market reactions, and reputational outcomes across heterogeneous institutional regimes (Candelon et al. 2021; De La Fuente and Velasco 2024; Kim and Yoo 2025). Taken together, these differences suggest that ESG decoupling is studied either in depth within specific institutional settings or more broadly across countries, but rarely in ways that fully integrate both perspectives.

Industry coverage further reveals an uneven focus on context. Multi-industry firm-level samples dominate (58.1%), supporting broad inference but potentially obscuring sector-specific dynamics (Chen et al. 2024a; Li et al. 2024a; Zhao et al. 2024). Financial institutions account for 14.5% of studies, reflecting heightened regulatory scrutiny and the role of finance in ESG signaling and discipline (Disli et al. 2022; Eliwa et al. 2023; Algeri et al. 2025). Research on environmentally sensitive industries (12.9%) highlights how environmental regulation can simultaneously constrain and incentivize symbolic ESG behavior (Tao et al. 2024; Tong et al. 2024; Tian et al. 2025; Yin et al. 2024), while sector-specific niche studies remain limited (8.0%) (Todaro and Torelli 2024; Fathoni et al. 2025). This uneven sectoral distribution indicates that while general patterns can be identified, important industry-specific mechanisms of ESG decoupling may remain underexplored.

Overall, these patterns indicate that existing knowledge on ESG decoupling is shaped by institutionally concentrated and sectorally uneven evidence, which strengthens internal validity in specific settings but limits cross-context generalizability.

3.3 | Characteristics (C)

Building on the theoretical perspectives reviewed above, this section synthesizes the ESG decoupling literature by organizing the determinants and consequences of ESG decoupling into a set of core mechanisms (see Figure 3 and Appendix A—Table A3). Rather than examining drivers and outcomes separately, the analysis groups empirical findings according to shared causal logics that explain why ESG decoupling arises and how it affects organizational and market outcomes. This mechanism-based structure integrates insights from governance, institutional, informational, and resource-based theories, enabling a comparative understanding of how different causal logics explain ESG decoupling as a process shaped by the interplay of organizational and external forces. The following subsections develop each mechanism by jointly discussing its key antecedents and resulting consequences.

3.3.1 | Agency and Governance Mechanisms

This mechanism explains ESG decoupling as a governance-driven outcome shaped by managerial self-interest, executive incentives, and the effectiveness of internal oversight structures.

When compensation schemes emphasize short-term rewards, executives are more likely to rely on symbolic ESG disclosures to secure reputational or organizational benefits without engaging in substantive sustainability actions (Li et al. 2024a; Li and Chen 2024).

Leadership dynamics further reinforce these incentives. CEO turnover and instability in top management teams are consistently associated with higher ESG decoupling, as newly appointed executives use ESG reporting as a rapid signaling device to establish legitimacy and manage external expectations during transitional periods (Niu et al. 2024; Yoo et al. 2024; Deng et al. 2025). These patterns highlight how executive discretion and career concerns influence ESG-related strategic choices.

Board-level governance mechanisms act as important internal constraints. Higher board activity and stronger board independence reduce ESG decoupling by strengthening monitoring and limiting managerial discretion (Disli et al. 2022). Board composition also matters. Greater board gender diversity and balanced executive gender representation are associated with more substantive ESG engagement, reflecting stronger ethical scrutiny and a more stakeholder-oriented governance process (Eliwa et al. 2023; Fathoni et al. 2025; Yang et al. 2024). In addition, executives with a stronger ESG orientation are less likely to engage in symbolic ESG practices (Wang et al. 2025a).

Conversely, certain governance configurations exacerbate ESG decoupling. Governance networks and interlocking directorships weaken effective oversight and facilitate the diffusion of symbolic ESG practices (Liu et al. 2024b), while political and bank-related governance ties further distort incentives and monitoring (Liu 2025). State-owned enterprises exhibit particularly persistent ESG decoupling due to competing political and commercial objectives that dilute accountability (Zhao et al. 2024; Chen et al. 2024a).

Governance-driven ESG decoupling has clear consequences. Internally, symbolic ESG engagement increases operating costs and labor inefficiency (Algeri et al. 2025; Di and Li 2023). Externally, it heightens reputational risk and undermines customer trust, leading to customer instability (Venturelli et al. 2024; Chen et al. 2024b).

Taken together, these findings indicate that governance-related factors primarily shape ESG decoupling through internal incentive structures and monitoring effectiveness, highlighting the central role of managerial discretion in driving symbolic ESG behaviour.

3.3.2 | Institutional and Regulatory Mechanisms

Beyond internal governance dynamics, this mechanism captures ESG decoupling in response to external regulatory pressures, institutional incentives, and political-economic conditions that shape firms' disclosure behavior. In such contexts, firms may prioritize regulatory conformity and external legitimacy over substantive changes in ESG performance.

Regulatory mandates constitute a central external driver. Mandatory ESG disclosure requirements are associated with lower ESG

decoupling when enforcement is credible and monitoring is effective (Aboud et al. 2024; Jin et al. 2024). However, regulatory interventions may also encourage short-term symbolic compliance, particularly in transitional settings or when enforcement is uneven. Similar mixed effects are observed in the context of green finance and environmental policies. While green finance initiatives, emissions trading schemes, and environmental regulations can constrain ESG decoupling, they may also incentivize strategic disclosure responses when firms focus on signaling compliance rather than implementing substantive ESG actions (Hu et al. 2025; Tong et al. 2024; Tao et al. 2024; Tian et al. 2025; Yin et al. 2024; Jiang et al. 2024a; Jiang et al. 2024b).

Political and market incentives further influence ESG decoupling. Firms' political contributions are associated with higher levels of ESG decoupling, as political engagement can be used to enhance perceived legitimacy and ESG ratings without corresponding improvements in sustainability performance (Carè et al. 2025). Investor behavior also plays an important role. Longer investor horizons are associated with lower ESG decoupling, reflecting stronger monitoring and greater emphasis on long-term sustainability outcomes (Eliwa and Elmaghrabi 2025). In contrast, mutual fund investor distraction weakens external discipline and allows ESG decoupling to persist (Liu et al. 2023).

Institutional-driven ESG decoupling has clear market-level consequences. In the short term, symbolic ESG compliance may be rewarded with lower debt and equity costs and positive market reactions (Wang et al. 2025; Peng and Xie 2024; Huang et al. 2024b). Over time, persistent ESG decoupling contributes to capital misallocation, undermines firm valuations, and increases reputational risk as inconsistencies between ESG disclosures and performance become apparent (Candelon et al. 2021; De La Fuente and Velasco 2024; Yoo et al. 2024; Venturelli et al. 2024).

Overall, these findings suggest that institutional and regulatory forces play a dual role, both constraining and enabling ESG decoupling, depending on the strength of enforcement and the incentives for symbolic compliance.

3.3.3 | Information Asymmetry and Signaling Mechanisms

Complementing both governance- and institution-based explanations, this mechanism frames ESG decoupling as a result of firms' ability to strategically manage ESG-related information and disclosures in environments characterized by information asymmetry. When transparency is limited or selectively controlled, firms can signal ESG commitment without undertaking corresponding substantive actions.

External information intermediaries play a key role in shaping these dynamics. Higher analyst coverage is associated with increased ESG decoupling, as firms under scrutiny may rely on symbolic ESG disclosures to manage expectations and maintain favorable market perceptions (Gao and Chen 2024). ESG ratings and disclosure incentives further reinforce this behavior, particularly when rating systems emphasize disclosure volume rather than performance quality, encouraging symbolic compliance (Liao et al. 2023; Li et al. 2025a). Investor attention also

affects ESG decoupling, with mixed effects depending on whether scrutiny strengthens accountability or amplifies short-term signaling incentives (Kim and Yoo 2025).

The information environment and technological infrastructure condition firms' disclosure strategies. Weakly regulated network infrastructures facilitate the diffusion of symbolic ESG information, enabling firms to project a positive external image without substantive internal change (Guan et al. 2024). In contrast, digital transformation reduces ESG decoupling by strengthening internal controls, improving data traceability, and enhancing transparency, thereby limiting firms' ability to rely on symbolic disclosure (Chen et al. 2024a; Sun et al. 2025a; Sun et al. 2025b; Xu et al. 2024). Information-driven ESG decoupling generates both short-term benefits and longer-term costs. In the short term, symbolic ESG signaling can trigger positive market reactions (Wang et al. 2025; Huang et al. 2024b). Over time, persistent decoupling undermines innovation by diverting resources away from substantive ESG initiatives (Zhang 2023; Sun et al. 2025b), erodes customer trust (Chen et al. 2024b), increases market mispricing (Lin et al. 2023), contributes to ESG rating divergence and lower liquidity (Chen et al. 2025), and heightens stock price crash risk (Liu et al. 2024c; Hussain et al. 2025).

These results highlight that ESG decoupling is not only driven by underlying incentives but also by how information is produced, mediated, and interpreted, emphasizing the role of disclosure systems and market perception in shaping symbolic behavior.

3.3.4 | Resource and Capability Mechanisms

Moving from external and informational drivers to internal constraints, this mechanism explains ESG decoupling as a consequence of firms' financial constraints, strategic orientations, and ability to mobilize resources for substantive sustainability action. When internal resources and capabilities are insufficient, firms may rely on symbolic ESG practices as a cost-efficient response to external expectations rather than engaging in genuine ESG integration.

Financial constraints represent a central internal driver of ESG decoupling. Firms facing limited financial flexibility are more likely to engage in symbolic ESG disclosure to enhance access to capital or improve external perceptions, without undertaking resource-intensive sustainability investments (Zhang 2022; Wang et al. 2025a). Strategic orientation further reinforces this pattern, as firms pursuing aggressive or short-term-focused business strategies tend to prioritize competitive positioning over substantive ESG implementation (Wang et al. 2025b).

External resource environments can mitigate these pressures. Digital finance inclusion reduces ESG decoupling by easing credit constraints and improving access to financial resources, thereby enabling firms to better align ESG commitments with actual performance (Li et al. 2024b, 2024a; Liu and Li 2024). Similarly, the adoption of circular economy practices supports more efficient resource use and facilitates the integration of sustainability into core operations, thereby reducing ESG decoupling (Todaro and Torelli 2024).

Resource-driven ESG decoupling has important consequences. Internally, persistent symbolic ESG engagement reduces investment in innovation and research and development, undermining long-term sustainability capabilities (Zhang 2023; Sun et al. 2025a). It also leads to operational inefficiencies and sub-optimal resource allocation (Algeri et al. 2025; Di and Li 2023; Kweh et al. 2026). Externally, ESG decoupling affects financing conditions, generating mixed effects on the cost of capital depending on market perceptions (Peng and Xie 2024; Wang et al. 2025; Huang et al. 2024b). In contrast, improved access to digital finance supports sustainability gains by strengthening firms' ability to translate ESG commitments into substantive outcomes (Sun et al. 2025b; Zhang 2023).

Taken together, these mechanisms illustrate that ESG decoupling is a multi-dimensional phenomenon shaped by the interaction of internal governance conditions, external institutional pressures, information environments, and resource constraints. Rather than representing a single causal pathway, ESG decoupling emerges from the alignment—or misalignment—of these forces, which may reinforce or counteract one another across different organizational and institutional contexts.

3.4 | Method (M)

The ESG decoupling literature is methodologically dominated by quantitative designs (see Appendix A—Table A4), with most studies relying on panel regressions to link ESG misalignment to firm-level characteristics and governance or institutional factors (Chen et al. 2024a; Li et al. 2024a; Niu et al. 2024; Zhao et al. 2024). A further stream employs quasi-experimental approaches, mainly difference-in-differences and policy-shock designs, to identify the effects of regulatory and institutional changes on ESG decoupling (Aboud et al. 2024; Jin et al. 2024; Tao et al. 2024; Tong et al. 2024; Tian et al. 2025). While these methods support causal inference, they largely frame ESG decoupling as an observable outcome rather than as an evolving organizational process.

More advanced econometric techniques, such as instrumental variables, propensity score matching, and mediation or moderation analyses, are used to address endogeneity and explore specific transmission channels (Peng and Xie 2024; Liu et al. 2024b; Long et al. 2025; Kim and Yoo 2025). Event-study designs focus on short-term market reactions to ESG disclosures, ratings, or controversies, prioritizing investor responses over internal dynamics (Candelon et al. 2021; Huang et al. 2024a; De La Fuente and Velasco 2024; Huang and Sun 2024). In contrast, qualitative and mixed-method studies are rare, providing only limited insight into how ESG decoupling is produced, justified, or sustained within organizations (Todaro and Torelli 2024; Kweh et al. 2026). Taken together, these methodological choices indicate that the literature is better equipped to explain the observable outcomes and market consequences of ESG decoupling than to capture its internal dynamics and underlying mechanisms.

Data choices reinforce this pattern. Most studies rely on commercial ESG databases, such as Refinitiv, MSCI, or Bloomberg, to operationalize ESG decoupling (Chen et al. 2024a; Li et al. 2024a; Eliwa et al. 2023), or on Chinese institutional databases,

reflecting the strong geographical focus of the literature (Niu et al. 2024; Jin et al. 2024; Tao et al. 2024). Because these sources often combine disclosure-based indicators (i.e., measures derived from firms' reported ESG policies and disclosures) with perception-based indicators (i.e., measures based on external assessments such as ESG ratings or stakeholder evaluations), they may obscure the distinction between symbolic ESG practices (focused on communication) and substantive ESG practices (reflecting actual organizational actions and performance). Hand-collected disclosures and alternative data sources remain limited, restricting opportunities to validate ESG claims against internal practices or stakeholder experiences (Venturelli et al. 2024).

Overall, existing methodological and data choices have generated robust evidence on correlates and consequences of ESG decoupling, but they provide limited insight into its internal dynamics and temporal evolution, highlighting the need for greater methodological diversity and richer data sources to capture ESG decoupling as an evolving organizational process.

4 | Developing an Integrative Framework for ESG Decoupling

Building on the systematic synthesis of determinants and consequences presented in the preceding section, this study develops an integrative conceptual framework that explains ESG decoupling as a dynamic organizational process shaped by the interplay of structural, behavioral, and institutional forces in which firms continuously adjust their ESG disclosure and practices in response to internal conditions, external pressures, and feedback from prior outcomes. This section provides an interpretative integration of the review findings, linking the identified mechanisms to broader theoretical explanations and clarifying their implications for understanding ESG decoupling as an evolving and context-dependent phenomenon.

Prior empirical studies often examine individual drivers or effects in isolation, offering limited insight into how organizational conditions, external pressures, and strategic responses interact over time. The framework proposed here moves beyond dominant static classifications—such as internal versus external determinants (Talpur et al. 2024; Velte 2023) or positive versus negative consequences (Seow 2023)—by consolidating fragmented evidence into a coherent structure that links underlying mechanisms, temporal dynamics, and feedback effects shaping symbolic and substantive ESG practices.

The framework is grounded in systems theory, the behavioral theory of the firm, and performance feedback theory. *Systems theory* conceptualizes organizations as embedded in interconnected internal and external environments in which outcomes arise through interaction and feedback rather than linear cause–effect relations (Bertalanffy 1968; Leoni 2024). In this framework, internal organizational factors—such as governance structures and resource capabilities—interact with external conditions, including regulatory environments and market pressures, while information systems operate at the firm–environment interface. ESG decoupling emerges from these dynamics, as internal responses are continuously shaped by external feedback and institutional constraints.

Behavioral and performance feedback theories explain how firms respond to discrepancies between actual performance and aspiration levels (Cyert and March 1992; Greve 2003). In ESG contexts, firms evaluate their sustainability performance relative to aspiration levels, creating performance–aspiration gaps that shape strategic responses. When ESG performance falls below aspiration levels, firms may engage in symbolic disclosure to protect legitimacy or avoid costly adjustments. Conversely, when performance meets or exceeds aspirations, firms may reinforce existing practices, either sustaining symbolic behavior or deepening substantive ESG engagement, depending on how feedback is interpreted (Seow 2023).

Together, these perspectives justify conceptualizing ESG decoupling as an adaptive process that may become self-reinforcing or self-correcting depending on how feedback is generated and interpreted (see Figure 4). Accordingly, the framework captures ESG decoupling as a system-level outcome arising from the interaction between internal organizational conditions and external institutional and informational environments. Beyond providing a conceptual foundation, the framework integrates previously fragmented theoretical perspectives by linking governance, institutional, informational, and resource-based mechanisms to their underlying theoretical lenses, thereby offering a unified explanation of ESG decoupling rather than isolated interpretations.

Building on these theoretical foundations and the empirical patterns identified in the review, the framework organizes ESG decoupling into four core mechanisms based on shared causal logics rather than isolated variables. This mechanism-based structure reflects the review's central insight that ESG decoupling cannot be adequately understood by separating drivers from outcomes. Governance arrangements, regulatory environments, information structures, and resource constraints not only trigger ESG decoupling but also shape its economic, reputational, and operational consequences, which in turn influence subsequent organizational behavior.

A central contribution of the framework lies in its explicit integration of temporal dynamics and feedback loops, extending the analysis beyond static explanations of ESG decoupling. In the short term, organizations may engage in tactical ESG decoupling, relying on symbolic disclosure to manage legitimacy pressures, investor expectations, or regulatory scrutiny. When such practices generate favorable market or reputational responses, reinforcing feedback may stabilize or intensify decoupling behavior. Over time, persistent misalignment between disclosure and performance can trigger corrective feedback through reputational damage, regulatory intervention, or financial constraints, prompting organizational adjustment or structural realignment (see Figure 4).

By distinguishing between reinforcing and corrective feedback processes, the framework explains why ESG decoupling may persist in some contexts while being attenuated in others. These feedback processes also clarify how ESG decoupling evolves from short-term discrepancies into long-term outcomes. When reinforcing feedback dominates, initial symbolic responses are reinforced and evolve into structural decoupling. In contrast, when corrective feedback prevails, early discrepancies trigger adjustments that realign disclosure with underlying

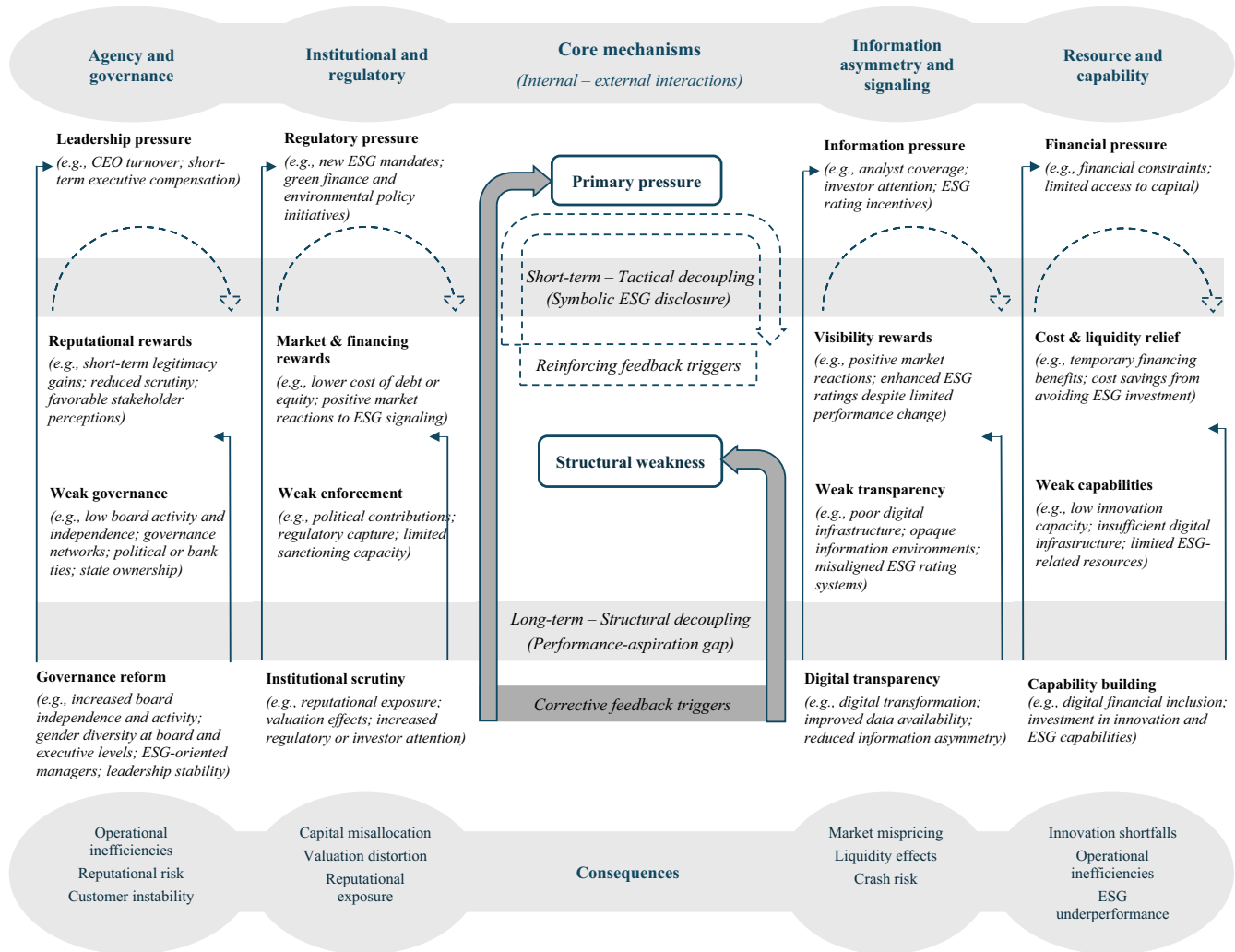


FIGURE 4 | ESG decoupling conceptual framework.

performance, reducing ESG decoupling over time. In this way, reinforcing and corrective dynamics represent alternative evolutionary paths rather than separate or contradictory states. Importantly, these feedback processes are recursive: the outcomes of ESG decoupling—such as reputational gains or penalties, financing conditions, and market reactions—feed back into managerial decision-making, shaping subsequent disclosure strategies and influencing whether firms continue, intensify, or correct symbolic ESG behavior over time.

This dynamic perspective advances existing static classifications of determinants and consequences by showing how ESG decoupling evolves over time through ongoing adjustments in response to feedback, rather than representing a fixed reporting outcome, and provides a foundation for examining ESG decoupling as an evolving organizational trajectory rather than a one-off reporting choice. An important insight emerging from this integrative perspective is the presence of tensions across mechanisms. For example, regulatory interventions and enhanced disclosure requirements are typically expected to reduce ESG decoupling; however, evidence suggests they may also incentivize symbolic compliance when enforcement is weak or compliance costs are high. Similarly, increased investor attention and analyst coverage can strengthen monitoring but may also amplify short-term

signaling incentives, thereby reinforcing decoupling behavior. These patterns suggest that mechanisms intended to enhance ESG accountability may produce unintended or counteracting effects depending on contextual conditions.

Within *agency and governance mechanisms*, ESG decoupling initially emerges as a short-term, tactical response to leadership changes and incentive misalignment. Newly appointed CEOs and unstable top management teams rely on symbolic ESG disclosure to manage first impressions and secure legitimacy (Niu et al. 2024; Yoo et al. 2024; Deng et al. 2025), particularly when executive compensation emphasizes short-term rewards (Li et al. 2024a; Li and Chen 2024). In the long term, ESG decoupling becomes structural when governance weaknesses persist and reinforcing feedback stabilizes initial symbolic responses over time. Limited board monitoring and independence reduce accountability (Disli et al. 2022), while governance networks, political or bank-related ties, and state ownership further embed ESG-performance gaps within organizational structures (Liu et al. 2024b; Liu 2025; Zhao et al. 2024; Chen et al. 2024a). Reinforcing feedback loops arise when short-term reputational and incentive-based rewards sustain symbolic ESG behavior in the absence of effective oversight (Li et al. 2024a; Li

and Chen 2024). Corrective feedback loops emerge when governance quality improves, as stronger board activity and independence (Disli et al. 2022), greater gender diversity at board and executive levels (Eliwa et al. 2023; Fathoni et al. 2025; Yang et al. 2024), leadership stability, and ESG-oriented managers constrain managerial discretion and enable a gradual transition from symbolic disclosure toward more substantive ESG alignment, thereby reducing ESG decoupling and influencing subsequent managerial decisions regarding ESG disclosure strategies (Wang et al. 2025a).

Within the *institutional and regulatory mechanism*, ESG decoupling initially manifests as a short-term, tactical response to new regulatory mandates and heightened institutional scrutiny. Firms adjust ESG disclosures to signal compliance following regulatory changes (Aboud et al. 2024; Jin et al. 2024) and in response to green finance or environmental policy initiatives (Hu et al. 2025; Tong et al. 2024; Tao et al. 2024), primarily to manage immediate legitimacy pressures. In the long term, ESG decoupling becomes structural in environments characterized by weak enforcement and political influence, as initial symbolic compliance responses are reinforced and persist over time under weak enforcement and limited accountability. Political contributions and institutional capture allow firms to sustain symbolic ESG practices without substantive change (Carè et al. 2025), while short-term investor horizons and mutual fund distraction weaken external monitoring and discipline (Liu et al. 2023; Yin et al. 2024; Jiang et al. 2024a; Jiang et al. 2024b). Reinforcing feedback loops arise when symbolic ESG compliance is rewarded through favorable market reactions (Huang et al. 2024b) or improved financing conditions (Wang et al. 2025; Peng and Xie 2024), delaying substantive ESG reform. Corrective feedback loops emerge when reputational exposure (Venturelli et al. 2024), capital misallocation (Candelon et al. 2021; De La Fuente and Velasco 2024), and valuation effects increase scrutiny and pressure firms to engage more authentically in ESG (Yoo et al. 2024).

In the context of *information asymmetry and signaling mechanisms*, ESG decoupling initially manifests as a short-term signaling strategy that exploits information gaps between firms and external stakeholders. Firms rely on symbolic ESG disclosure to enhance visibility and manage investor perceptions (Gao and Chen 2024), particularly under analyst scrutiny and ESG rating incentives (Liao et al. 2023; Li, Mao, et al. 2025b), generating favorable short-term market reactions (Wang et al. 2025; Huang et al. 2024b). In the long term, ESG decoupling persists in information-poor environments, as symbolic signaling remains effective in shaping external perceptions without being corrected by transparent information flows, particularly in contexts characterized by weak transparency and poorly governed digital infrastructures (Guan et al. 2024), as well as misaligned ESG rating systems that fail to penalize symbolic disclosure (Chen et al. 2025). Reinforcing feedback loops arise when positive market responses reward superficial ESG signaling (Wang et al. 2025), encouraging the continuation of symbolic disclosure strategies (Gao and Chen 2024). Corrective feedback loops emerge when digital transformation initiatives reduce information asymmetries (Chen et al. 2024a; Sun et al. 2025a; Sun et al. 2025b; Xu et al. 2024), allowing market penalties—such as mispricing (Lin et al. 2023), liquidity effects, and rating divergence (Chen

et al. 2025), or stock price crash risk (Liu et al. 2024c; Hussain et al. 2025)—to constrain ESG decoupling over time.

Lastly, in the *resource and capability mechanism*, ESG decoupling initially emerges as a short-term survival strategy driven by financial constraints. Firms facing limited access to capital rely on symbolic ESG disclosure to secure investor confidence or alleviate financing pressure (Zhang 2022), particularly when strategic orientations emphasize short-term performance over capability development (Wang et al. 2025b). In the long term, ESG decoupling becomes structural when firms lack the organizational capabilities, innovation capacity, or internal resources required to implement substantive ESG practices, as symbolic signaling remains effective in shaping external perceptions without being corrected by transparent information flows. Persistent resource constraints reduce investment in ESG-related processes and lead to operational inefficiencies, thereby embedding ESG performance gaps over time (Algeri et al. 2025; Di and Li 2023). Reinforcing feedback loops arise when symbolic ESG practices generate temporary financial relief or market support, thereby encouraging firms to maintain superficial engagement rather than invest in costly capability-building (Zhang 2022). Corrective feedback loops emerge when firms enhance their resource base through digital financial inclusion and capability development, thereby reducing financial constraints and enabling a gradual transition toward more substantive ESG engagement (Li et al. 2024b; Liu et al. 2024c; Sun et al. 2025b).

Overall, the framework provides a theory-driven explanation of ESG decoupling as an adaptive process. By organizing evidence around core mechanisms and explicitly incorporating temporal patterns and feedback effects, it clarifies when ESG disclosure misalignment is likely to persist and when it becomes more amenable to correction. From a practical perspective, the findings suggest that improving ESG disclosure alone is insufficient to ensure substantive sustainability outcomes, highlighting the need for stronger governance structures, credible regulatory enforcement, and improved transparency mechanisms. From a research perspective, the framework highlights the need for more process-oriented and context-sensitive studies, particularly those that capture the dynamic, multi-level nature of ESG decoupling.

5 | Future Research Directions

Building on the systematic review and synthesis of 62 empirical studies, this section presents a comprehensive research agenda for ESG decoupling. Following the structure proposed by Paul and Rosado-Serrano (2019), we utilize the TCCM (Theory–Context–Characteristics–Methodology) framework to identify key gaps and guide future scholarly work. The suggested directions are organized into four interconnected dimensions—emphasizing underexplored theoretical perspectives, contextual factors, explanatory mechanisms, and methodological approaches. Although the recommendations are based on explicit suggestions from previous studies, they are not exhaustive. Researchers are encouraged to critically evaluate and adapt these directions to the evolving landscape of ESG practices, regulations, and disclosures. A summary of the proposed directions and related research questions is provided in Figure 5.

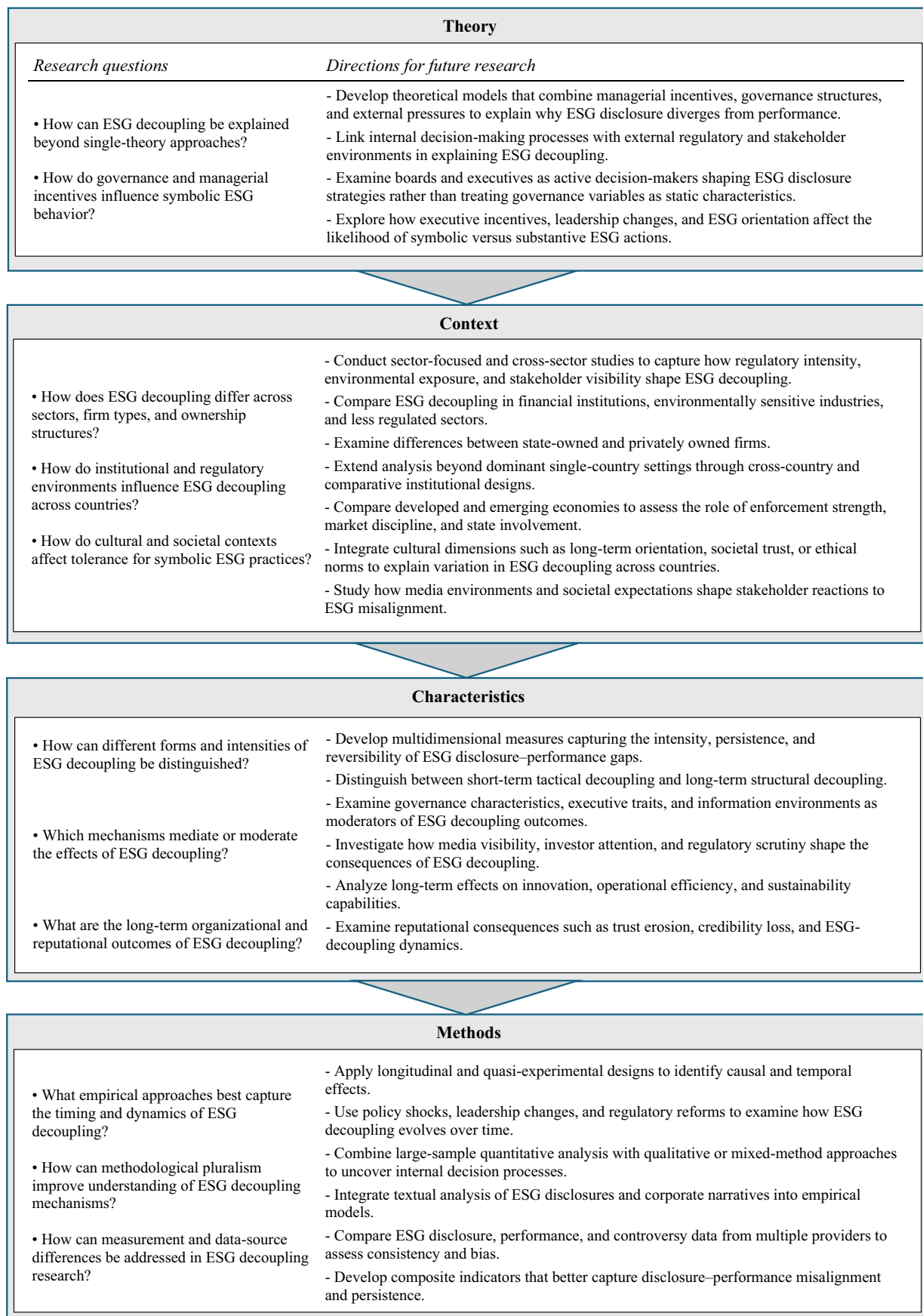


FIGURE 5 | Future research.

5.1 | Theory (T)

Future research on ESG decoupling should move beyond the isolated or loosely connected use of theoretical perspectives and develop integrated, mechanism-oriented theoretical models. Governance-based theories require refinement by conceptualizing boards and executive incentives as active channels of influence, rather than static attributes. Board activity, independence, and diversity should be examined as mechanisms that shape managerial discretion and symbolic disclosure strategies, including their mediating and moderating roles in ESG decoupling (Disli et al. 2022; Eliwa et al. 2023; Fathoni et al. 2025). Extending this line of inquiry also requires incorporating broader board characteristics—such as tenure, expertise, and network embeddedness—to better explain variation in monitoring effectiveness (Yang et al. 2024; Liu et al. 2024b).

The literature further calls for stronger theoretical integration. Agency-based explanations of managerial self-interest need to be combined with institutional and legitimacy perspectives to capture how internal incentives interact with external regulatory and stakeholder pressures in sustaining ESG decoupling (Li et al. 2024a; Aboud et al. 2024; Venturelli et al. 2024). Similarly, signaling and information asymmetry theories should be embedded within governance and institutional frameworks to explain why symbolic ESG practices persist despite increasing disclosure and transparency requirements (Candelon et al. 2021; Liao et al. 2023; Chen et al. 2024a).

Advancing theory also requires a dynamic perspective. Several studies highlight the need for process-based theorization that explains how ESG decoupling emerges, stabilizes, or reverses over time in response to feedback from markets, regulators, and stakeholders, rather than treating decoupling as a static outcome (Niu et al. 2024; Zhao et al. 2024; Wang et al. 2025).

Finally, future work should strengthen micro–macro linkages by connecting managerial cognition, risk preferences, and short-termism with organizational and institutional outcomes. Behavioral and upper-echelons perspectives highlight the importance of individual-level drivers, yet these remain poorly integrated into broader explanations of ESG decoupling (Li and Chen 2024; Lin et al. 2023; Kweh et al. 2026).

5.2 | Context (C)

Future research on ESG decoupling should expand its contextual scope by addressing the strong geographic and institutional concentration of existing evidence. Much of the literature is based on single-country settings—most notably China—where regulatory transitions, state ownership, and governance reforms play a central role. While analytically valuable, this concentration limits generalizability. Comparative studies across developed and emerging economies are needed to examine how differences in regulatory enforcement, market discipline, and state involvement shape ESG decoupling dynamics (Li et al. 2024a; Zhao et al. 2024; Kim and Yoo 2025).

A second priority concerns sectoral context. The dominance of multi-industry samples may obscure sector-specific mechanisms

through which ESG decoupling operates. Future research should adopt sector-focused or cross-sector comparative designs to capture how regulatory intensity, environmental exposure, and stakeholder visibility condition symbolic ESG behavior. This is particularly relevant for financial institutions and environmentally sensitive industries, where ESG incentives and constraints differ markedly from those in less regulated sectors (Disli et al. 2022; Algeri et al. 2025; Tao et al. 2024).

Finally, ESG decoupling research would benefit from incorporating cultural and societal contexts. Existing studies rarely examine how cultural norms and social expectations influence firms' tolerance for symbolic ESG practices or stakeholders' responses to misalignment. Integrating cultural dimensions—such as long-term orientation or societal trust—offers a promising avenue for explaining cross-country variation in reputational sanctions and the persistence of ESG decoupling (Haidar 2025; Venturelli et al. 2024).

5.3 | Characteristics (C)

Future studies should refine the characterization of ESG decoupling by distinguishing between short-term tactical misalignment and long-term structural decoupling. Several papers call for measures that capture not only the presence of ESG decoupling but also its intensity, persistence, and reversibility, particularly in settings shaped by leadership changes, ownership structures, or regulatory transitions (Niu et al. 2024; Zhao et al. 2024; Chen et al. 2024a).

Future research should also examine how specific drivers lead to different consequences of ESG decoupling. Rather than estimating average effects, studies are encouraged to trace whether governance- or institution-driven decoupling primarily affects reputational outcomes, market valuation, operational efficiency, or innovation capacity over time (Candelon et al. 2021; Zhang 2023; Liu et al. 2024c).

Finally, future work should focus on when and how ESG decoupling is corrected. Authors highlight the need to identify conditions under which misalignment becomes visible and costly—such as heightened transparency, regulatory tightening, or changes in investor monitoring—and to identify the triggers that prompt realignment between ESG disclosure and performance (Venturelli et al. 2024; Chen et al. 2024b).

5.4 | Method (M)

Future research on ESG decoupling should strengthen methodological rigor by moving beyond predominantly cross-sectional and linear empirical designs. First, studies should increasingly adopt longitudinal and quasi-experimental approaches to better identify causal relationships and dynamic effects. Difference-in-differences designs, staggered regulatory shocks, and event-based analyses are particularly well suited to capturing how ESG decoupling emerges, intensifies, or corrects following changes in regulation, leadership, or market scrutiny (Aboud et al. 2024; Jin et al. 2024; Tao et al. 2024; Tian et al. 2025).

Second, future studies should address endogeneity and selection concerns more systematically. Several papers point to the need for advanced econometric techniques—such as instrumental variables, propensity score matching, or dynamic panel models—to disentangle strategic ESG behavior from underlying firm characteristics and unobserved heterogeneity (Peng and Xie 2024; Liu et al. 2024b; Wang et al. 2025a). Without such approaches, estimates of ESG decoupling risk conflate symbolic behavior with structural differences across firms.

Third, methodological progress requires greater attention to measurement and construct validity. Authors call for improved operationalization of ESG decoupling that goes beyond single-score gaps by incorporating multiple ESG dimensions, disclosure–performance mismatches, and temporal persistence (Chen et al. 2024a; Zhao et al. 2024). Combining ESG ratings with firm-level performance indicators or controversy data would provide a more robust assessment of misalignment.

Finally, future research would benefit from methodological pluralism. While quantitative panel studies dominate the field, mixed-method and qualitative designs remain rare but valuable for unpacking internal decision processes and governance dynamics underlying ESG decoupling (Todaro and Torelli 2024; Kweh et al. 2025). Integrating qualitative insights with large-sample evidence would enhance the explanatory depth of research on ESG decoupling.

6 | Conclusions

This study provides the first systematic, theory-driven synthesis of ESG decoupling research, with a specific focus on its determinants, consequences, and the dynamic feedback processes that sustain or correct disclosure–performance misalignment. Based on 62 empirical articles published between 2021 and 2025, the review followed the SPAR-4-SLR protocol to ensure transparency and replicability and employed the TCCM framework to organize and evaluate evidence across theories, contexts, core characteristics, and methodological approaches.

Building on this structured synthesis, the study develops an integrative conceptual framework that explains ESG decoupling through four mechanism clusters—agency and governance, institutional and regulatory, information asymmetry and signaling, and resource and capability—and conceptualizes decoupling as an adaptive organizational process shaped by temporal patterns (short-term tactical decoupling versus long-term structural misalignment) and reinforcing or corrective feedback loops from markets and stakeholders.

The study makes three interrelated contributions to the ESG and sustainability reporting literature. *Conceptually*, it advances prior research by shifting the analytical focus from static, fragmented explanations toward a process-oriented, mechanism-based understanding of ESG decoupling. By integrating diverse theoretical perspectives into a unified framework, the study clarifies how governance, institutional, informational, and resource-related forces jointly shape the emergence, persistence, and correction of ESG misalignment.

Empirically, the review provides a comprehensive synthesis of recent evidence across contexts and methodologies, highlighting systematic patterns in how ESG decoupling manifests under different regulatory, market, and organizational conditions. This integrative perspective extends existing literature by moving beyond isolated findings and offering a coherent explanation of when symbolic ESG practices are likely to persist or be corrected.

Methodologically, the study contributes by systematically organizing a rapidly expanding empirical literature using the SPAR-4-SLR protocol and the TCCM framework, thereby enhancing transparency, comparability, and the development of cumulative knowledge. In doing so, it identifies important gaps in current research designs—particularly the limited use of longitudinal, process-oriented, and mixed-method approaches—highlighting opportunities for future empirical advancement.

More broadly, the study contributes to ongoing debates about the credibility and effectiveness of ESG reporting by demonstrating that disclosure cannot be evaluated in isolation from underlying organizational dynamics and feedback processes. By positioning ESG decoupling as a dynamic and systemic phenomenon, the study provides a stronger foundation for understanding the limitations of current reporting regimes and for designing more effective governance and regulatory interventions.

The findings have important implications for policymakers, corporate managers, and market participants concerned with the credibility and effectiveness of ESG reporting. *For policymakers and regulators*, the mechanism-based framework suggests that disclosure mandates alone are insufficient to ensure substantive sustainability outcomes. Rather, the effectiveness of ESG regulation depends on the quality of enforcement, assurance mechanisms, and the alignment between reporting requirements and underlying performance indicators. In this respect, regulatory interventions should be designed not only to increase the quantity of disclosure but also to strengthen verification, comparability, and accountability, thereby reducing the scope for symbolic compliance.

For corporate managers and boards, the analysis shows that ESG decoupling is not merely a reporting problem but may reflect deeper governance and capability misalignments. Strengthening board oversight, aligning executive incentives with long-term ESG objectives, and investing in internal data systems and monitoring processes are important conditions for translating ESG commitments into substantive practices. Firms should also recognize that short-term reputational gains from symbolic disclosure may lead to longer-term operational and financial risks, reinforcing the need for integrated ESG strategies that link disclosure more closely to actual performance.

For investors and civil society, the findings highlight the limitations of relying on disclosure-based ESG metrics in isolation. The framework suggests that external stakeholders should place greater emphasis on the consistency between ESG disclosure and performance indicators, as well as on longitudinal patterns of ESG behavior. Active engagement, longer investment horizons, and closer scrutiny of ESG ratings and assurance practices may

help discourage symbolic ESG behavior and promote more substantive sustainability outcomes.

Several *limitations* should be acknowledged. First, the review focuses on peer-reviewed empirical studies published between 2021 and 2025, thereby excluding earlier conceptual contributions and insights from practice-oriented or regulatory sources. Second, much of the ESG decoupling literature relies on secondary ESG databases and quantitative designs, limiting insight into internal organizational processes, managerial decision-making, and the evolution of decoupling over longer time horizons. Third, although systematic coding guided the synthesis, the classification of determinants, consequences, and theoretical perspectives inevitably involves interpretive judgment.

These limitations point to promising directions for *future research*. Building on the framework developed in this study, future work should focus on identifying the conditions under which ESG decoupling emerges, persists, or is corrected over time, particularly by examining how governance, institutional, informational, and resource-based mechanisms interact. Further research would benefit from longitudinal and process-oriented designs that capture how ESG practices evolve in response to market and regulatory feedback. In addition, the use of alternative data sources—such as assurance information, ESG controversies, or qualitative evidence—can help better distinguish between symbolic disclosure and substantive performance. Finally, comparative studies across countries, industries, and regulatory contexts would provide a deeper understanding of how ESG decoupling varies across settings and when corrective mechanisms are most effective.

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Appendix A

TABLE A1 | Theories used in ESG decoupling research.

Theory	Key reasons for their use	Frequency (# studies)	Sources
Agency	Explains ESG decoupling as a consequence of managerial self-interest, short-term incentives, and weak monitoring	16	Li et al. (2024a); Disli et al. (2022); Deng et al. (2025); Yang et al. (2024); Yoo et al. (2024); Zhao et al. (2024); Liu (2025)
Stakeholder	Firms symbolically disclose ESG information to appease stakeholders without incurring substantive costs	12	Algeri et al. 2025; Disli et al. 2022; Fathoni et al. 2025; Li et al. 2024a; Li et al. 2024b; Venturelli et al. 2024; Wang et al. 2025
Signaling	ESG disclosure is used to send positive signals to markets even when performance does not match	10	Candelon et al. 2021; Liao et al. 2023; Tao et al. 2024; Zhang and Xiong 2025; Venturelli et al. 2024; Xu et al. 2024
Institutional	Firms symbolically conform to regulatory and normative pressures under weak or ambiguous enforcement	8	Aboud et al. 2024; Disli et al. 2022; Eliwa et al. 2023; Li et al. 2024a; Tao et al. 2024; Wang et al. 2025b
Upper echelons	Executive traits (gender, experience, turnover) shape symbolic vs. substantive ESG choices	8	Eliwa et al. 2023; Fathoni et al. 2025; Wang et al. 2025a; Xu et al. 2024; Yang et al. 2024; Niu et al. 2024
Resource dependence	Firms rely on symbolic ESG disclosure to secure capital, political support, or legitimacy	7	Carè et al. 2025; Disli et al. 2022; Liu et al. 2024b; Sun et al. 2025b; Tao et al. 2024
Legitimacy	ESG disclosure is used to preserve social acceptance when performance lags	7	Aboud et al. 2024; Chen et al. 2025; Haidar 2025; Sun et al. 2025b; Venturelli et al. 2024; Yin et al. 2024
Information asymmetry	Managers exploit information gaps to selectively disclose favorable ESG information	7	Chen et al. (2024a), Lin et al. (2023), Liu et al. (2024a), Sun et al. (2025b), Wang et al. (2025b)
Impression management	ESG reporting is crafted to shape stakeholder perceptions rather than reflect reality	6	Guan et al. 2024; Hussain et al. 2025; Liao et al. 2023; Lin et al. 2023; Liu et al. 2024b; Long et al. 2025
Behavioral	Short-termism and performance pressure drive symbolic ESG responses	4	Lin et al. 2023; Liu et al. 2024a; Li and Chen 2024; Liu et al. 2024a
Organizational reputation	Symbolic ESG is used to protect or restore reputation under scrutiny	4	Chen et al. 2024a; Chen et al. 2024b; Tong et al. 2024; Zhang and Xiong 2025
Voluntary disclosure	Because reporting is voluntary, firms strategically disclose positive ESG aspects while withholding negatives.	2	Liu et al. 2024a; Zhang 2022
Resource-based view	Symbolic ESG disclosure is treated as a resource to build competitiveness without developing true capabilities.	2	Zhang and Xiong 2025; Kweh et al. 2025
Prospect	Under risk or performance decline, managers prefer low-cost symbolic disclosure to avoid losses rather than substantive action.	2	Kweh et al. 2025; Lin et al. 2023
Other theories (e.g., gender socialization, gone fraud, neo-pluralist)		25	Carè et al. 2025; Yang et al. 2024; Yin et al. 2024; Peng and Xie 2024; Wang et al. 2025; Haidar 2025; Huang et al. 2024a
No theory		15	Di and Li 2023; Eliwa and Elmaghrabi 2025; Gao and Chen 2024; Hu et al. 2025; Jin et al. 2024; Kim and Yoo 2025; Qu 2024; Tian et al. 2025; Todaro and Torelli 2024

TABLE A2 | Context settings used in ESG decoupling research.

First-level	Sub-level	Frequency (%)	Representative references
Single-country studies	China	59.7%	Chen et al. (2024a); Li et al. (2024a); Niu et al. (2024); Zhao et al. (2024); Wang et al. (2025a)
	Other single-country contexts (US, Brazil, Korea, Turkey, Pakistan)	8.0%	Carè et al. (2025); Deng et al. (2025); Yoo et al. (2024); Kweh et al. (2025); Hussain et al. (2025)
Regional studies	Europe/EU (developed economies)	8.1%	Disli et al. (2022); Eliwa et al. (2023); Aboud et al. (2024); Algeri et al. (2025); Venturelli et al. (2024)
	Other regional settings (emerging-markets)	1.6%	Disli et al. (2022)
Global/multi-country studies	Cross-country/global	22.6%	Candelon et al. (2021); De La Fuente and Velasco (2024); Venturelli et al. (2024); Kim and Yoo (2025); Todaro and Torelli (2024)

Region	Countries covered	Frequency (%)	Representative references
Asia	China; Korea; Pakistan; Turkey; ASEAN countries	69.4%	Chen et al. (2024a); Niu et al. (2024); Zhao et al. (2024); Yoo et al. (2024); Hussain et al. (2025); Fathoni et al. (2025)
Europe	EU member states; pan-European samples	8.1%	Disli et al. (2022); Eliwa et al. (2023); Aboud et al. (2024); Algeri et al. (2025); Venturelli et al. (2024)
Americas	United States; Brazil	4.8%	Carè et al. (2025); Deng et al. (2025)
Global/multi-continent	Multiple continents; international firm samples	17.7%	Candelon et al. (2021); De La Fuente and Velasco (2024); Kim and Yoo (2025); Todaro and Torelli (2024); Venturelli et al. (2024)

Industry focus	Frequency (%)	Representative references
Multi-industry/economy-wide samples	58.1%	Chen et al. (2024a); Li et al. (2024a); Niu et al. (2024); Zhao et al. (2024); Wang et al. (2025a)
Financial institutions (banks & financial firms)	14.5%	Disli et al. (2022); Eliwa et al. (2023); Aboud et al. (2024); Algeri et al. (2025); Eliwa and Elmaghrabi (2025)
Environmentally sensitive/polluting industries	12.9%	Hu et al. (2025); Tao et al. (2024); Tian et al. (2025); Tong et al. (2024); Yin et al. (2024)
Manufacturing-focused samples	6.5%	Di and Li (2023); Zhang (2023); Xu et al. (2024); Sun et al. (2025b)
Sector-specific niche contexts (e.g., agri-food, circular economy)	8.0%	Todaro and Torelli (2024); Kweh et al. (2025); Fathoni et al. (2025)

TABLE A3 | Characteristics of ESG decoupling.

	Theoretical foundation	Determinants	Consequences
Agency and governance mechanisms	— Agency — Stakeholder — Upper Echelons — Organizational reputation — Gender socialization	• CEO compensation (I/–) (Li et al. 2024a; Li and Chen 2024) • CEO turnover (I/–) (Niu et al. 2024; Yoo et al. 2024; Deng et al. 2025) • Board activity (I/+) (Disli et al. 2022) • Board independence (I/+) (Disli et al. 2022) • Gender diversity (I/+) (Eliwa et al. 2023; Fathoni et al. 2025) • Executive gender composition (I/+) (Yang et al. 2024) • Managerial ESG orientation (I/+) (Wang et al. 2025a) • Governance networks/interlocks (I/–) (Liu et al. 2024b) • Political/bank governance ties (I/–) (Liu 2025) • SOEs governance (I/–) (Zhao et al. 2024; Chen et al. 2024a)	• Operating costs (I/–) (Algeri et al. 2025) • Labor inefficiency (I/–) (Di and Li 2023) • Reputational risk (E/–) (Venturelli et al. 2024) • Customer instability (E/–) (Chen et al. 2024b)
Institutional and regulatory mechanisms	— Institutional — Neo-Institutional — Legitimacy — Public interest — Neo-pluralist	• Regulatory mandates (E/+) (Aboud et al. 2024; Jin et al. 2024) • Green finance/environmental policy (E/±) (Hu et al. 2025; Tong et al. 2024; Tao et al. 2024; Tian et al. 2025; Yin et al. 2024; Jiang et al. 2024a; Jiang et al. 2024b) • Political contributions (E/–) (Carè et al. 2025) • Investor horizons (E/+) (Eliwa and Elmaghrabi 2025) • Mutual fund investor distraction (E/–) (Liu et al. 2023)	• Cost of debt and equity yields (E/+) (Wang et al. 2025; Peng and Xie 2024) • Market reaction (E/+) (Huang et al. 2024b) • Fund misallocation (E/–) (Candelon et al. 2021; De La Fuente and Velasco 2024) • Firm valuation (E/+) (De La Fuente and Velasco 2024; Yoo et al. 2024) • Reputational risks (E/–) (Venturelli et al. 2024)
Information asymmetry and signaling mechanisms	— Signaling — Information asymmetry — Impression management — Voluntary disclosure — Gone fraud	• Digital transformation (I/+) (Chen et al. 2024a; Sun et al. 2025b; Xu et al. 2024) • Analyst coverage (E/–) (Gao and Chen 2024) • Network infrastructure (E/–) (Guan et al. 2024) • Investor attention (E/±) (Kim and Yoo 2025) • ESG ratings/disclosure incentives (E/–) (Liao et al. 2023; Li et al. 2025a)	• Market reactions (E/+) (Wang et al. 2025; Huang et al. 2024b) • Reduced innovation (I/–) (Zhang 2023; Sun et al. 2025a) • Customer trust (E/+) (Chen et al. 2024b) • Market mispricing (E/–) (Lin et al. 2023) • ESG rating divergence; liquidity effects (E/–) (Chen et al. 2025) • Stock price crash risk (E/–) (Liu et al. 2024c; Hussain et al. 2025)
Resource and capability mechanisms	— Resource-based view — Resource dependence — Behavioral — Prospect — Credit rationing — Classical micro-finance	• Financial constraints (I/–) (Zhang 2022; Wang et al. 2025a) • Digital finance inclusion (E/+) (Li et al. 2024b; Liu et al. 2024a; Liu and Li 2024) • Business strategy (I/–) (Wang et al. 2025b) • Circular economy practices (E/+) (Todaro and Torelli 2024)	• Innovation and R&D reduction (I/–) (Zhang 2023; Sun et al. 2025a) • Financing conditions (E/±) (Peng and Xie 2024; Wang et al. 2025; Huang et al. 2024b) • Operational inefficiencies (I/–) (Algeri et al. 2025; Di and Li 2023; Kweh et al. 2025) • Sustainability gains via digital finance (E/+) (Sun et al. 2025b; Zhang 2023)

Note: Internal (I); external (E); positive (+); negative (–).

TABLE A4 | Methods used in ESG decoupling research.

Methodological category (% of total)	Typical technique	Frequency (# studies)	Representative references
Quantitative panel regressions (45.2%)	Fixed-effects panel regression	21	Chen et al. (2024a); Li et al. (2024a); Niu et al. (2024); Zhao et al. (2024)
	Random-effects panel regression	7	Wang et al. (2025a); Yang et al. (2024)
	OLS with firm and year fixed effects	12	Zhang (2022); Li et al. (2024a); Wang et al. (2025a)
Quasi-experimental designs (22.6%)	Difference-in-differences (DiD)	9	Aboud et al. (2024); Jin et al. (2024); Tao et al. (2024)
	Policy shock/regulatory reform design	5	Tong et al. (2024); Tian et al. (2025)
Advanced econometric methods (16.1%)	Instrumental variables (IV)	4	Peng and Xie (2024); Liu et al. (2024b)
	Propensity score matching (PSM)	3	Wang et al. (2025a); Kim and Yoo (2025)
	Dynamic panel models (GMM)	2	Liu et al. (2024b); Zhang (2023)
	Mediation/moderation analysis	6	Wang et al. (2025a); Kim and Yoo (2025)
Event-study methodologies (8.1%)	Short-window market reaction studies	5	Candelon et al. (2021); Huang et al. (2024b); De La Fuente and Velasco (2024)
Qualitative or mixed methods (8%)	Mixed qualitative–quantitative design	5	Venturelli et al. (2024); Todaro and Torelli (2024); Kweh et al. (2025)
Data source category (% of total)	Sub-category	Frequency (# studies)	Representative references
Commercial ESG databases (43.5%)	Refinitiv ESG/ASSET4	15	Chen et al. (2024a); Li et al. (2024a); Zhao et al. (2024); Wang et al. (2025a)
	MSCI ESG/KLD	6	Candelon et al. (2021); Deng et al. (2025); Kim and Yoo (2025)
	Bloomberg ESG	6	Eliwa et al. (2023); Venturelli et al. (2024); De La Fuente and Velasco (2024)
Chinese ESG and institutional databases (29.0%)	CSMAR/WIND/HEXUN	18	Niu et al. (2024); Jin et al. (2024); Tao et al. (2024); Tong et al. (2024); Tian et al. (2025)
Hand-collected ESG disclosures (11.3%)	Annual reports/sustainability reports/websites	7	Venturelli et al. (2024); Todaro and Torelli (2024); Fathoni et al. (2025)
Financial and market databases (stand-alone) (9.7%)	Compustat/CRSP/Datastream	6	Peng and Xie (2024); De La Fuente and Velasco (2024)
Survey or alternative data (6.5%)	Questionnaires/proprietary or experimental data	4	Kweh et al. (2025); Hussain et al. (2025)

Appendix B

TABLE B1 | Sample details.

Authors	Theory	Sample	Period	Methodology	Data sources	Main results
Aboud et al. 2024	Legitimacy, institutional	EU Listed firms + US control group	2010–2020	Difference-in-differences (DiD) + Propensity Score Matching (PSM)	Refinitiv ESG; EU Directive 2014/95 regulatory data	EU Directive 2014/95 mitigates ESG decoupling; effect stronger in non-controversial industries; weaker when ESG information is externally assured; national enforcement strength not significant.
Algeri et al. 2025	Stakeholder	Europe listed banks	2006–2021	Stochastic Frontier Analysis (SFA)	Banks' financial & ESG reports; Executive remuneration data	Introducing ESG in the cost frontier significantly affects banks' cost function; stronger ESG orientation improves long-term cost efficiency; ESG-linked executive pay associates with lower efficiency.
Cadelon et al. 2021	Signaling	International equity mutual funds	2000–2019	Panel regression; Comparative analysis (signals vs. extra-financial ratings)	Morningstar; MSCI; Novethic; Fund ESG labels	Evidence of fund-level ESG-washing (labels not aligned with holdings); underperformance and transparency issues motivate regulatory standardization.
Carè et al. 2025	Resource dependence, neo-pluralist	US Listed firms	2010–2023	System GMM (Arellano-Bond dynamic panel estimation)	LSEG Workspace (Refinitiv/Thomson Reuters)	Political contributions raise E and G scores and are linked to ESG controversies; limited effect on S pillar.
Cepêda et al. 2025b	Agency	International listed firms	2009–2023	Panel regression + interaction (long-term vs. short-term investors)	Refinitiv; Global firm-level databases	Institutional investors help reduce ESG decoupling; effect depends on investment horizon (long-term vs. short-term).
Chen et al. 2025	Agency, legitimacy, stakeholder, signaling	China Listed firms	2015–2020	Residual method for ESG washing + PSM + Entropy balancing + DiD	Huazheng; SynTao; CASVI; Wind ESG rating agencies	Greater ESG washing increases rating divergence (especially environmental); divergence reduces stock liquidity; stronger with more private info and lower rater accuracy.
Chen et al. (2024a)	Organizational reputation, information asymmetry	China A-share listed firms	2010–2019	Text analysis + Panel regression (fixed effects)	A-share ESG reports; GRI adoption indicators	Corporate digital transformation reduces ESG decoupling via better information processing and lower information asymmetry; stronger in eastern regions and non-GRI firms.
Chen et al. (2024b)	Organizational reputation, information asymmetry	China A-share listed firms + Top 5 customers	2010–2019	Panel regression + Mediation	Manually collected ESG reports; Customer-supplier data	ESG decoupling lowers customer stability; mechanisms: reduced reputation and increased information asymmetry; stronger in SOEs, non-high-tech firms, and competitive industries.
Deng et al. 2025	Agency	US Publicly traded firms	2010–2018	Panel regression + Moderation	US ESG ratings; Compustat financial data	ESG disparity weakens the value effect of ESG; effect intensifies with agency problems and lower monitoring.
De La Fuente and Velasco 2024	Agency, stakeholder	China A-share listed firms	2009–2022	OLS + PSM + Mechanism analysis	CSMAR (China Stock Market & Accounting Research Database)	Top management team stability reduces ESG greenwashing; mechanisms: lower agency costs and higher disclosure quality; stronger in non-SOEs and with non-Big Four auditors.
Di and Li 2023	No theory	China A-share listed non-financial firms	2013–2021	Panel regression (OLS, firm and year FE) + Entropy balancing robustness	Bloomberg (ESG disclosure), Huazheng (ESG performance), CSMAR (financials), WIND (labor data)	ESG decoupling reduces labor investment efficiency, increasing both over- and under-investment; effect stronger under environmental uncertainty.
Disli et al. 2022	Institutional, resource dependence, stakeholder, agency	Emerging markets listed non-financial firms	2010–2019	Dynamic panel (two-step System GMM) + year and industry FE	Refinitiv ESG scores; Thomson Reuters Eikon (board attributes, financials)	Smaller, more independent and gender-diverse boards improve sustainability performance; frequent board meetings reduce ESG controversies.
Eliwa and Elmaghrabi (2025)	No theory	Global Listed firms	2005–2023	Panel regression	Global ownership and firm-level datasets	Long-term institutional investors reduce ESG decoupling; short-term investors show no significant effect.

TABLE B1 | (Continued)

Authors	Theory	Sample	Period	Methodology	Data sources	Main results
Eliwa et al. (2023)	Upper echelons, gender socialization, institutional	Europe Listed firms	2005–2019	Panel regression + Instrumental variables (IV) + PSM robustness	Refinitiv ESG/Eikon (disclosure and performance), World Values Survey (religiosity)	Board gender diversity reduces ESG decoupling; religiosity weakens this effect, especially in highly religious contexts.
Fathoni et al. (2025)	Stakeholder, upper echelons, agency, biosocial constructionist	Global publicly-listed banks	2011–2023	Panel regressions + Fixed effects (bank, year, country) + IV (2SLS)	Bloomberg (ESG disclosure), Refinitiv Eikon (ESG performance, bank data), World Bank (macro data)	ESG controversies increase ESG-washing; women on boards mitigate washing, with a critical-mass effect (~21.5%).
Gao and Chen (2024)	No theory	China A-share listed firms	2012–2022	Panel regressions + IV + PSM robustness + industry/year FE	Bloomberg (ESG disclosure), Huazheng (ESG performance), CSMAR (analyst coverage, fundamentals)	Analyst coverage increases ESG greenwashing (analyst pressure hypothesis); effect stronger for high-reputation, poor-performing firms; star analysts mitigate.
Guan et al. (2024)	Impression management	China A-share listed firms in Broadband China pilot vs. non-pilot cities	2010–2022	Difference-in-differences (DID) quasi-natural experiment	Firm-level ESG/financial data merged with Broadband China city lists	Broadband China increases ESG greenwashing; effect stronger with higher financial constraints, mitigated by greater analyst coverage.
Haidar (2025)	Attribution, social norm, legitimacy, moral licensing	International Multi-country listed firms	2002–2021	Cross-country panel regressions + FE	ESG controversy scores; country-level trust indices	Higher societal trust reduces greenwashing; effect stronger in high-visibility firms and weaker with strong formal institutions.
Hu et al. (2025)	No theory	China Firms in Green Finance Reform & Innovation Pilot Zones	2011–2023	Difference-in-differences (DID) with mechanism/heterogeneity tests	Chinese firm-level ESG/financial datasets + policy zone lists	Green finance reform increases ESG greenwashing; strong internal controls and digitalization mitigate.
Huang and Sun (2024)	Agency	China A-share listed firms	2012–2021	Difference-in-differences (DID) using Social Credit System variation	Firm-level ESG and financial data merged with SCS indices	Favorable social credit environment reduces ESG greenwashing; effect stronger under high agency costs and poor disclosure quality.
Huang et al. (2024a)	Corporate governance, institutional	China A-share listed firms	2012–2022	Panel regression + 2SLS	CSMAR; Bloomberg/Refinitiv	Renewable energy technology innovation reduces ESG greenwashing; effect stronger with board experiential diversity and higher media attention.
Huang et al. (2024b)	Agency, stakeholder	International Banks (syndicated loans)	1999–2016	Panel regression + Event study	DealScan; Refinitiv MarketPsych	Banks with worse ESG lend more to better-ESG firms, offering favorable loan terms; markets react positively; effect stronger in polluting industries.
Hussain et al. (2025)	Agency, impression management	USA & China Listed firms	2008–2022	Fixed-effects panel regressions	Refinitiv Datastream	ESG decoupling increases stock price crash risk in China (especially non-SOEs) but not in the USA; investor sentiment does not moderate the effect.
Jiang et al. (2024a)	No theory	China GEM vs. Main Board firms	2012–2022	Difference-in-differences (DID) + IPW	CSMAR; ESG ratings	Registration-based IPO reform increases ESG greenwashing; peer competition and retail sentiment mediate; effect stronger in polluting industries and low-transparency firms.
Jiang et al. (2024b)	No theory	China A-share listed firms	2015–2021	Difference-in-differences (DID) + PSM-DID	Bloomberg; CSMAR	Environmental Protection Tax increases ESG greenwashing (~13%); effect stronger in large firms and developed regions with strong enforcement.
Jin et al. (2024)	No theory	China Listed firms	2015–2019	Policy-shock panel analysis	Bloomberg; Huazheng	Green financial regulation shifts firms toward disclosure over practice, potentially increasing ESG greenwashing; heterogeneity by executive backgrounds and firm perceptions.

(Continues)

TABLE B1 | (Continued)

Authors	Theory	Sample	Period	Methodology	Data sources	Main results
Kim and Yoo (2025)	No theory	Global Public firms	2016–2021	Panel regression (signaling model)	Google SVI; Bloomberg	Firms adjust ESG washing in response to retail investor attention; increased attention reduces washing when perceived as scrutiny.
Kweh et al. (2025)	Stakeholder, prospect, resource-based view	Global Microsoft supply chain firms	2011–2020	Two-stage DEA + regression	Firm-level ESG and controversy indicators	Overall ESG improves firm efficiency; controversies reduce efficiency; E and G pillars are positive, S and controversies negative.
Li and Chen (2024)	Tournament, behavioral, relative exploitation, risk sensitivity, self-attribution, social comparison	China A-share listed firms	2011–2021	Panel regression + mechanism tests	Bloomberg/Huazheng; CSMAR	Wider executive pay gaps increase ESG greenwashing via risk-taking; effect stronger with low environmental awareness, weak regulation, and high financing constraints.
Li et al. (2024a)	Information asymmetry, agency, upper echelons, stakeholder, institutional	China A-share listed firms	2013–2022	Fixed-effects panel regressions + mechanism tests	CSMAR; ESG ratings	Higher CEO cash pay increases greenwashing; equity incentives reduce it; internal control mitigates; Big-Four auditing and media coverage amplify cash-pay effects; environmental regulation strengthens equity-pay inhibition.
Li et al. (2024b)	Stakeholder, signaling	China A-share listed firms	2012–2022	Fixed-effects panel regressions + mediation	CSMAR; ESG ratings	Digital transformation inhibits ESG greenwashing; internal control and financing constraints partially mediate; stronger in non-high-pollution, high-tech, and eastern regions.
Li et al. (2024c)	No theory	China A-share listed firms	2011–2022	Fixed-effects panel regressions + IV	Digital Financial Inclusion Index; CSMAR; ESG ratings	Digital financial inclusion reduces ESG greenwashing via easing financing constraints and lowering information asymmetry; effects stronger with lower analyst coverage and higher government supervision.
Li et al. (2025b)	Efficient market, limited attention, spotlight, legitimacy, stakeholder, managerial myopia, signaling	China A-share listed firms	2011–2022	Fixed-effects panel regressions + IV + moderation	SSE/SZSE e-interaction; CSMAR; ESG ratings	Retail attention increases greenwashing by boosting disclosure without improving performance; effect stronger with financing constraints and weaker governance; professional investor attention mitigates.
Liao et al. (2023)	Deterrence, impression management, signaling	China A-share listed firms	2015–2021	OLS + Fixed-effects panel regressions	CSRC comment letters; CSMAR; ESG ratings	Greenwashing increases following financial report comment letters; stronger with higher letter intensity; better governance reduces the response.
Lin et al. (2023)	Behavioral finance, information asymmetry, impression management, prospect	China A-share listed firms	2011–2021	Fixed-effects panel regressions	CSMAR; Bloomberg/Asset4; WIND	ESG greenwashing increases equity mispricing (especially overpricing); mitigated by high-quality auditing, stronger competition, and stricter environmental regulation; peer greenwashing raises mispricing.
Liu and Li (2024)	No theory	China A-share listed firms and banks	2011–2022	Fixed-effects panel regressions + mediation	Bank fintech metrics; CSMAR; ESG ratings	Bank fintech reduces greenwashing by easing financing constraints and information asymmetry; stronger in high-polluting firms and those without bank shareholding or SOE status.
Liu et al. (2023)	No theory	China A-share listed firms	2017–2018	Fixed-effects panel regressions	Mutual fund distraction measures; CSMAR; ESG ratings	Distracted institutional investors exacerbate ESG decoupling; firms face less monitoring pressure when investors are distracted.

(Continues)

TABLE B1 | (Continued)

Authors	Theory	Sample	Period	Methodology	Data sources	Main results
Liu et al. (2024a)	Behavioral finance, behavioral theory of the firm, resource dependence, signaling, voluntary disclosure, credit rationing, information asymmetry, upper echelons	China A-share listed firms	2016–2022	Fixed-effects panel regressions + mediation	Digital Finance Index; CSMAR; ESG ratings	Digital finance suppresses ESG decoupling via alleviating financing constraints, improving disclosure, increasing investment efficiency, and reducing managerial myopia; effects stronger in non-SOEs, high-tech, and heavy-polluting firms; investor attention positively moderates.
Liu et al. (2024b)	Resource dependence, agency, impression management, agency	China A-share listed firms	2014–2020	Panel regression (firm & year FE) + DID + Heckman + 2SLS IV	CSMAR; Bloomberg; Wind & Huazheng	Vertical interlock increases ESG report greenwashing and conceals negative ESG; results robust across DID/Heckman/IV.
Liu et al. (2024c)	No theory	China A-share listed firms	2014–2021	Panel regression (industry & year FE) + PSM + robustness checks	Bloomberg; Refinitiv; CSMAR	ESG report greenwashing is associated with higher stock price crash risk; effect stronger in firms with high earnings management.
Liu (2025)	Institutional, agency	China A-share listed firms	2011–2020	Panel regression (firm & year FE) + robustness tests	CSMAR; Bloomberg; Wind ESG; Huazheng	Bank shareholding and executives with banking backgrounds increase greenwashing; transparency reduces it; stronger in non-SOEs and weaker financial markets.
Long et al. (2025)	Agency, impression management	China A-share listed firms	2011–2021	Panel regression (firm & year FE) + IV (2SLS) + Heckman two-stage	East Money message boards sentiment; Bloomberg; China Securities ESG; CSMAR; Wind	Negative social-media sentiment induces ESG greenwashing through managerial myopia; mitigated by institutional ownership, high-quality audit, and stronger regulation.
Niu et al. (2024)	Upper-echelons	China A-share listed firms	2009–2022	Panel regression (industry & year FE) + PSM + 2SLS IV + Tobit + Heckman two-stage	Bloomberg; Refinitiv; Wind ESG; CSMAR	CEO turnover increases ESG greenwashing—especially external successors and under higher performance pressure.
Peng and Xie (2024)	Financing cost	China listed firms	2011–2021	Panel regression (FE) + FRL robust estimation	CSMAR, CNRDS	Corporate ESG greenwashing raises debt financing costs, especially for non-SOEs and smaller firms.
Qu (2024)	No theory	China listed firms	2009–2022	Panel regression (FE) + robustness checks	Bloomberg; Wind ESG; CSMAR	Higher litigation risk curbs ESG greenwashing; D&O insurance fosters it by reducing this risk.
Sun et al. (2025a)	Legitimacy, information asymmetry, resource dependence	China A-share listed firms	2012–2021	Panel regression (FE) + mediation/moderation tests	Bloomberg; Wind ESG; Huazheng; CSMAR	Digital transformation reduces greenwashing via innovation, information, and resource channels; effect stronger with investor attention.
Tao et al. (2024)	Signaling, institutional, resource dependence	China A-share listed firms	2015–2021	Difference-in-differences (policy shock: ETS/EPT) + robustness	Bloomberg ESG disclosure; Wind ESG; CNRDS; CSMAR	China's ETS curbs ESG greenwashing, though effect varies with firm size, governance quality, and pollution intensity.
Tian et al. (2025)	No theory	China A-share listed firms	2014–2021	Difference-in-differences (MSCI inclusion) + robustness	CSMAR; Bloomberg; Wind ESG; Huazheng	Capital market internationalization (MSCI inclusion) reduces ESG greenwashing by alleviating financing constraints and enhancing external oversight.
Todaro and Torelli (2024)	No theory	International high-impact industries	2020–2021	Panel regression	ESG performance datasets	ESG-washing extends beyond environmental greenwashing in circular economy contexts; boundaries between substantive and symbolic practices are often blurred.
Tong et al. (2024)	Reputation	China A-share listed firms (pilot vs. non-pilot zones)	2012–2020	Difference-in-differences (DID) + robustness checks	CSMAR; CNRDS	Green finance pilot zones reduce ESG decoupling/greenwashing.

(Continues)

TABLE B1 | (Continued)

Authors	Theory	Sample	Period	Methodology	Data sources	Main results
Venturelli et al. (2024)	Legitimacy, stakeholder, signaling	International Banks	2011–2021	Panel regression (fixed effects)	Refinitiv Eikon ESG; BankScope; World Bank indicators	ESG-washing increases reputational exposure for banks, especially under stricter ESG regulation.
Wang et al. (2024)	GONE fraud, stakeholder	China Manufacturing & polluting listed firms	2011–2021	Panel regression (two-way fixed effects)	Bloomberg ESG; CSI ESG; CSMAR; CNRDS	Executives with environmental experience reduce brown washing; effect amplified by media attention and analyst coverage; limited effect on greenwashing/decoupling.
Wang et al. (2025a)	Upper echelons	China A-share listed firms	2010–2022	Panel regression (fixed effects) + mediation/moderation	Bloomberg ESG; Refinitiv Asset4; CSMAR	Prospector strategies increase ESG greenwashing through reputation and profit-seeking channels; institutional investors mitigate the effect.
Wang et al. (2025b)	Institutional, information asymmetry, resource dependence	China A-share listed firms	2012–2021	Panel regression (fixed effects) + mechanism analysis	Bloomberg ESG; Refinitiv Asset4; CSMAR; CNRDS	Greenwashing lowers debt costs and raises stock returns in the short term; effects dissipate over longer horizons.
Xu et al. (2024)	Signal transmission, strategic cognition, upper echelons	China A-share listed firms	2012–2022	Panel regression (fixed effects) + mediation/heterogeneity	CSMAR; CNRDS	Digital transformation suppresses greenwashing/decoupling; stronger with environmentally aware executives and in SOEs.
Yang et al. (2024)	Agency, upper echelons, gender socialization	China listed firms	2011–2021	Panel regression (fixed effects) + 2SLS robustness	CSMAR; ESG ratings databases	Female executives reduce ESG decoupling; effect stronger in non-SOEs.
Yin et al. (2024)	Externality, neoclassical economic, legitimacy	China A-share listed polluting firms	2011–2020	Difference-in-differences (DID) + PSM-DID	CSMAR; regulatory filings	Environmental Protection Law reduces greenwashing; effect stronger in non-SOEs and developed regions.
Yoo et al. (2024)	Agency, value-enhancing	Korea listed firms	2012–2020	Logit + panel regressions; PSM/Heckman/IV robustness	KCGS ESG ratings; KIS-VALUE financials	Frequent CEO turnover increases ESG-washing as myopic behavior; ESG-washing reduces firm value.
Zhang and Xiong (2025)	Signaling, resource dependence, reputation	Global Firms	2002–2018	Panel regression (fixed effects) + robustness tests	Bloomberg ESG; Refinitiv ESG; Worldscope	Financially constrained firms are more likely to greenwash; financial intermediation alleviates this; leverage increases pressure to greenwash.
Zhang (2022)	Neoclassical, voluntary disclosure, trade-off	China A-share listed firms	2013–2019	Panel regression (fixed effects) + mediation tests	CNRDS patents; Bloomberg ESG; Refinitiv ESG; Huazheng ESG; CSMAR	Digital finance empowerment promotes green innovation by reducing extreme ESG hypocrisy; effect stronger in non-SOEs, polluting firms, and competitive industries.
Zhang (2023)	Classical micro-finance	China A-share listed agri-food firms	2011–2022	Panel regression + PSM + system GMM	Bloomberg ESG disclosure; Huazheng ESG ratings; CNRDS patents; CSMAR	ESG decoupling reduces innovation; mitigated by independent directors; stronger in large firms, food industry, and highly digitalized firms.
Zhao et al. (2024)	Agency	China Non-financial SOEs vs. non-SOEs	2011–2021	Difference-in-differences (DID) + PSM-DID + placebo tests	Bloomberg ESG disclosure; Sino-Securities ESG ratings; CSMAR; CNRDS	Classified Reform of SOEs alleviates ESG decoupling by reducing aspiration–performance gap; stronger in special-function SOEs; weakened by political connections and media attention.